

Household Demand Dynamics and Equity Mispricing

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Abstract

Household demand shocks to individual stocks generate larger valuation distortions when the institutional investors holding the stock have lower capacity to absorb the shocks. I measure stock-level absorption capacity as the inverse of the average price elasticity of demand among institutions holding the stock before the shock occurs, and find that limited institutional absorption capacity amplifies the price impact of otherwise similar household demand shocks. Specifically, a household demand shock generates a larger contemporaneous and subsequent price impact when the stock is held primarily by institutions with lower price elasticity of demand. The resulting mispricing is economically meaningful: stocks with higher household demand shocks, and lower absorption capacity earn abnormal returns of about 1.36% quarterly and about 3.90% annually, and this price impact persists for up to three years with a non-monotonically declining term structure.

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Introduction

Retail investors—referred to as households throughout this paper¹—frequently reallocate substantial amounts of capital across individual stocks. Because these demand shifts are usually unrelated to firm fundamentals (Koijsen & Yogo, 2019), they can deviate prices from fundamental values and generate meaningful mispricing. Besides their large magnitude, household demand for stocks is highly persistent (Huebner, 2023b; Van der Beck, 2026). That is, households usually become interested in certain stocks, and their interest only gradually reverses.²

This persistent feature of household demand for stocks has two pricing effects. First, the initial household demand shock generates demand pressure and a jump in prices. Conceptually, this price impact is stronger when the institutional investors holding the stock have less elastic capital to absorb it (Duarte et al., 2025). Second, as household demand gradually reverses, prices also reverse over time. Therefore, following a positive household demand shock, contemporaneous prices jump, and the subsequent reversal depends on the reversal of the underlying demand itself.

Figure 1 summarizes the dynamics of household demand for stocks and the associated post-shock price reversal. The green series, measured on the right axis, traces the latent household demand for a portfolio of stocks that experience high household demand in quarter t , highlighting its gradual reversal over the subsequent quarters. The black series, measured on the left axis, shows the quarterly expected returns associated with changes in household demand shocks. Stocks with higher latent household demand outperform their peers by about 1.2 percent in the subsequent quarter, and it takes about three years for the reversal to dissipate, following a non-monotonic decline in expected returns in absolute value. This result highlights the role of persistent household demand in shaping long-term prices.

¹Following the convention in the literature, stock-level household position is measured as the residual of institutional investors' reported 13F holdings.

²Such nonfundamental interest is usually interpreted as investor sentiment. However, household demand may also reflect idiosyncratic preferences for individual stocks. Therefore, this paper remains agnostic about the underlying motivation behind household demand shifts.

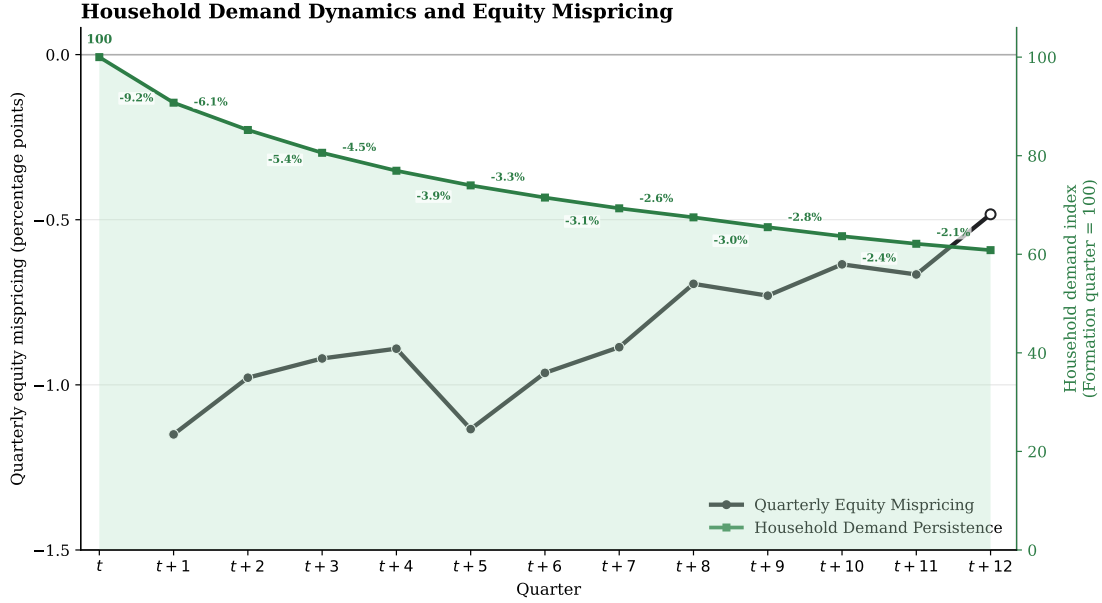


Figure 1: Household Demand Dynamics and Equity Mispricing. The black series, measured on the left axis, shows the estimated future quarterly return effects of household demand shocks. A value of -1 at the one-quarter horizon indicates that firms with a one-standard-deviation higher household demand shocks underperform their peers about %1 over the subsequent quarter. The green series, measured on the right axis, traces the persistence of household demand shocks, which traces the household latent demand of a portfolio of stock which experienced the highest household demand at quarter t .

To further differentiate between the two components of household demand, I use an AR(1) process to decompose household demand into an innovation component, which captures the new arrival of household demand for stocks, and an expected component, which captures the gradual reversal in household demand. The results show that the innovation component is associated with sharp underperformance in the subsequent quarter, while the expected component is associated with a long-term reversal, driving the long-term price impact in Figure 1. In other words, new household demand arrivals push contemporaneous prices sharply, generating a short-term reversal in the next quarter, while the gradual reversal in household demand governs the long-term underperformance of stocks.

The price impact of a household demand shock should depend not only on the size of the shock but also on the capacity of institutional investors to absorb it. This is consistent with theoretical and empirical findings in market microstructure (Bessembinder & Seguin,

1992, 1993; Chordia et al., 2002a; Cont et al., 2014a; Kyle, 1985a), which show that the price impact of order-flow imbalances depends on market depth, proxied by liquidity, open interest, and related measures. This paper defines the price elasticity of demand of institutional investors holding the stock before the shock hits as a measure of institutional investors' capacity to absorb the shock.

The mechanism is simple: consider the same household demand shock hitting two otherwise identical stocks, one predominantly held by pension funds—which adjust their holdings little in response to short-term price movements—and the other by hedge funds, which have more arbitrage capital for short-term opportunities. The shock generates a smaller price distortion in the stock held by hedge funds because a smaller price adjustment is needed to draw their more elastic arbitrage capital. Thus, otherwise identical household demand shocks can generate different valuation distortions depending on the elastic institutional capital available to absorb them.

To quantify these valuation distortions and the role of institutional absorption capacity, I construct a *Price Impact Multiplier*. The multiplier is the inverse of a stock's institutional absorption capacity and measures the model-implied price adjustment required for institutions to absorb a one-unit household demand shock. I then scale each household demand shock by the corresponding multiplier, allowing shocks of the same magnitude to have different predicted price effects depending on the elasticity of the institutional capital available to absorb them. Empirically, the multiplier amplifies the price effects of household demand shocks of the same magnitude. Conditional on the underlying shock, a higher predicted price impact is associated with a larger contemporaneous price increase and a stronger subsequent reversal that unfolds over approximately three years.

To test whether these price effects reflect mispricing rather than compensation for risk, I form portfolios based on the predicted price impact of household demand shocks. A strategy that buys low-impact stocks and shorts high-impact stocks earns six-factor abnormal returns of approximately 1.36% over the subsequent quarter and 3.90% over the subsequent year. The

return spread survives controls for firm size, idiosyncratic volatility, analyst coverage, and conventional limits-to-arbitrage proxies. Thus, institutional absorption capacity captures more than households' tendency to trade small or difficult-to-arbitrage stocks: the pricing consequences of household demand depend critically on the amount of elastic institutional capital available to absorb it.

Taken together, the findings highlight the importance of household demand dynamics for both short- and long-term prices. Household demand behaves like a state variable: innovations in household demand generate contemporaneous stock-price jumps, and these jumps are sharper when institutional investors have lower capacity to absorb the shock. At the same time, households' persistent interest creates long-lived mispricing that takes years to dissipate.

The remainder of the paper is organized as follows. Section 1 reviews the related literature. Section 2 develops the conceptual framework and its empirical predictions. Section 3 describes the data. Section 4 presents the main empirical results. Section 5 perform robustness tests. Section 6 concludes.

1 Related Literature

This paper first contributes to the demand-based asset-pricing literature pioneered by Kojen and Yogo (2019). Their framework uses portfolio holdings to estimate investor-specific demand elasticities and latent demand, thereby connecting shifts in investor demand to equilibrium asset prices. Subsequent work applies this approach across different investors and asset classes and studies how mandates, constraints, and investor heterogeneity affect valuations (Benetton & Compiani, 2020; Haddad et al., 2021; Han et al., 2021; Kojen & Yogo, 2020, 2024; Kojen et al., 2021). This literature establishes that asset demand is substantially less elastic than standard frictionless models imply and that investor demand can consequently have economically important price effects.

I extend this literature in three respects. First, I distinguish household demand from aggregate institutional demand and test its price impact and subsequent mispricing. The findings show that the predictability documented in Kojen and Yogo (2019), which is driven by aggregate latent demand, comes from household demand, while institutional latent demand has no return predictability; therefore, disaggregation improves price-impact estimation. Second, I characterize both the contemporaneous price response and the quarter-by-quarter return response over several years, showing that long-term mispricing is as economically important as the next-quarter reversal. While the literature typically assumes that latent demand reverts to its mean after one month and measures expected returns based on this assumption, I show that persistent household demand drives prices over the long term. Third, and most importantly, I estimate the role of institutional investors' limited absorption capacity and show that it amplifies the price effect of otherwise equivalent household demand shocks.

The paper is particularly related to recent dynamic extensions of asset-demand systems. Huebner (2023a) shows that return dynamics depend on both the persistence of demand shocks and the term structure of investor demand elasticities. Persistent demand shocks can sustain price effects, while subsequent mean reversion contributes to return reversal. Van der

Beck (2026) shows that investors become more elastic over longer horizons because portfolio adjustment is gradual. Consequently, the price impact of a demand shift is larger in the short run than in the long-run equilibrium. I provide complementary reduced-form evidence by separating newly arriving household demand from persistent household positions. Household innovations generate sharp short-run price pressure, whereas persistent household demand produces a non-monotonic return reversal lasting several years.

The second related literature is market microstructure. In Kyle (1985b), signed order flow moves prices through a price-impact coefficient that is inversely related to market depth. Information-based models interpret the permanent component of price impact as reflecting information revealed through trading (Glosten & Milgrom, 1985; Hasbrouck, 1991). Inventory and price-pressure models instead emphasize temporary price movements required to induce other investors to absorb trading imbalances (Bessembinder & Seguin, 1993; Scholes, 1972). Empirically, order-flow imbalances are strongly related to contemporaneous returns, and their price effects depend on available liquidity and market depth (Chordia et al., 2002b; Cont et al., 2014b). Trader identity also matters because households, institutions, and liquidity providers differ in their information and trading motives (Boehmer & Wu, 2013; Grinblatt & Keloharju, 2000).

I bring this microstructure intuition to a lower-frequency asset-pricing setting. Household-demand innovations provide a quarterly analogue of order-flow shocks, while institutional absorption capacity provides a stock-level analogue of market depth. Unlike quoted depth, institutional absorption capacity measures the quantity and elasticity of the institutional capital available to absorb household demand. Consistent with the microstructure evidence, a given household innovation produces a larger contemporaneous price increase when absorption capacity is limited. The subsequent reversal indicates that part of this initial price response represents temporary demand pressure rather than permanent information. The paper therefore connects transaction-level theories of order flow and depth to persistent portfolio imbalances observed in quarterly holdings.

Finally, the paper contributes to the literature on household trading and investor sentiment. Noise-trader models show that non-fundamental demand can move prices away from fundamental value and that limited arbitrage may allow such distortions to persist (De Long et al., 1990a). Empirical research finds that sentiment has particularly strong effects among stocks that are difficult to value or arbitrage (Baker & Wurgler, 2006; Stambaugh et al., 2012). Household investors are also especially responsive to attention and salient information when selecting individual stocks (Barber & Odean, 2008). These findings motivate examining whether household demand creates temporary price pressure rather than conveying permanent information about firm value.

My contribution is to connect household sentiment to an observable demand and market-clearing mechanism. Rather than relying on an aggregate sentiment index, I construct a stock-level household latent-demand measure from portfolio holdings, separate newly arriving demand from persistent positions, and measure the institutional capital available to absorb each shock. Household innovations generate immediate price pressure, while persistent household positions keep the imbalance embedded in prices and predict a gradual reversal. Institutional absorption capacity determines the intensity of the initial response. The evidence therefore identifies when household sentiment affects prices, how long its effects survive, and why otherwise similar household-demand shocks produce different pricing outcomes across stocks.

2 Theoretical Framework

This section develops a simple market-clearing framework linking household demand shocks, institutional absorption capacity, and equity mispricing. The framework has two investor sectors: households, denoted by H , and institutions, denoted by I . Stocks are indexed by j , and time is indexed by t . Let $p_{j,t}$ denote the log market price of stock j , and let $p_{j,t}^*$ denote the benchmark price that would prevail in the absence of excess demand. The valuation wedge, or distortion in fundamental driven price, therefore is

$$m_{j,t} \equiv p_{j,t} - p_{j,t}^*. \quad (1)$$

2.1 Excess Demand and Market Clearing

For investor sector $g \in \{H, I\}$, let $ED_{j,t}^g$ denote excess demand for stock j , measured as a fraction of the stock's shares outstanding. Excess demand is the difference between the sector's desired ownership and the ownership implied by prices and observed firm characteristics. It may reflect idiosyncratic preferences, private information, sentiment, or other unobserved determinants of portfolio demand.

Around the benchmark price, the change in sector g 's demand can be approximated by

$$\Delta q_{j,t}^g \simeq ED_{j,t}^g - \kappa_{j,t}^g m_{j,t}, \quad (2)$$

where $\kappa_{j,t}^g > 0$ measures the sector's absorption capacity. Because demand slopes downward, a positive price adjustment reduces the quantity demanded by $\kappa_{j,t}^g m_{j,t}$.

With fixed stock supply, market clearing requires

$$\Delta q_{j,t}^H + \Delta q_{j,t}^I = 0. \quad (3)$$

Substituting equation (2) into equation (3) gives

$$m_{j,t} = \frac{ED_{j,t}^H + ED_{j,t}^I}{\kappa_{j,t}^H + \kappa_{j,t}^I}. \quad (4)$$

Defining aggregate excess demand as

$$AED_{j,t} \equiv ED_{j,t}^H + ED_{j,t}^I, \quad (5)$$

the equilibrium valuation wedge becomes

$$m_{j,t} = \frac{AED_{j,t}}{\kappa_{j,t}^M}, \quad \kappa_{j,t}^M \equiv \kappa_{j,t}^H + \kappa_{j,t}^I. \quad (6)$$

Equation (6) shows that excess demand affects prices relative to fundamentals. Positive aggregate excess demand raises the market price above its benchmark value, while negative aggregate excess demand lowers it. The magnitude of this valuation wedge depends inversely on the market's aggregate absorption capacity.

2.2 Institutional Absorption Capacity

The paper focuses on household demand shocks and the institutional investors that accommodate them. Let $s_{i,j,t-1}$ denote institution i 's ownership of stock j as a fraction of the stock's shares outstanding, and let $\eta_{i,t-1} > 0$ denote the magnitude of institution i 's estimated demand elasticity. A local price change therefore induces an approximate change in the institution's demand equal to

$$\Delta q_{i,j,t}^I \simeq -s_{i,j,t-1} \eta_{i,t-1} \Delta p_{j,t}. \quad (7)$$

Aggregating across the institutions holding stock j gives institutional absorption capacity:

$$\kappa_{j,t-1}^I = \sum_{i \in \mathcal{I}_{j,t-1}} s_{i,j,t-1} \eta_{i,t-1}. \quad (8)$$

Equivalently,

$$\kappa_{j,t-1}^I = IO_{j,t-1} \bar{\eta}_{j,t-1}^I, \quad (9)$$

where $IO_{j,t-1}$ is total institutional ownership and $\bar{\eta}_{j,t-1}^I$ is the ownership-weighted average demand elasticity of the stock's institutional holders. Institutional absorption capacity is therefore high when institutions own a substantial fraction of the stock and their demand responds elastically to price.

Consider a household demand shock, $\nu_{j,t}^H$, expressed as a fraction of shares outstanding. If the shock is predetermined when it arrives and institutions supply the marginal offsetting position, market clearing requires

$$\nu_{j,t}^H - \kappa_{j,t-1}^I \Delta p_{j,t} = 0. \quad (10)$$

Consequently,

$$\Delta p_{j,t} = \frac{\nu_{j,t}^H}{\kappa_{j,t-1}^I}. \quad (11)$$

This expression is a short-run microstructure approximation to the full equilibrium in equation (4). It treats newly arriving household demand as the initiating order imbalance and institutions as the marginal absorbing sector. The empirical analysis also considers broader capacity measures as robustness, but institutional capacity is the central object because the paper asks how institutional capital accommodates household demand shocks.

2.3 The Price Impact Multiplier

Equation (11) motivates the *Price Impact Multiplier*:

$$\text{PIM}_{j,t-1} \equiv \left(\kappa_{j,t-1}^I \right)^{-1}. \quad (12)$$

A high Price Impact Multiplier identifies a stock with limited institutional absorption capacity. It measures the model-implied price adjustment required for institutions to absorb a one-unit household demand shock.

I define the *household-demand-implied price impact* as

$$\text{IPI}_{j,t}^H \equiv \text{PIM}_{j,t-1} \nu_{j,t}^H. \quad (13)$$

The implied price impact combines the magnitude of newly arriving household demand with the institutional capital available to absorb it. Holding the household shock constant, implied price impact is larger when institutional ownership is lower, when the stock’s institutional holders are less elastic, or both.

In the empirical implementation, I regularize the multiplier by imposing a quarter-specific fifth-percentile floor on institutional absorption capacity:

$$\text{PIM}_{j,t-1}^{\text{reg}} = \left[\max \left\{ \kappa_{j,t-1}^I, Q_{0.05,t-1}(\kappa^I) \right\} \right]^{-1}. \quad (14)$$

This regularization prevents near-zero estimated capacity from mechanically generating extreme multiplier values.

2.4 Demand Persistence and Future Returns

Household excess demand is persistent. To capture this feature, suppose that the household-demand state follows

$$HED_{j,t} = \rho_j^H HED_{j,t-1} + \nu_{j,t}^H, \quad 0 < \rho_j^H < 1, \quad (15)$$

where $\nu_{j,t}^H$ is the newly arriving household demand shock. Conditional on the current innovation, its expected contribution to household excess demand h quarters later is

$$E_t[HED_{j,t+h} \mid \nu_{j,t}^H] = (\rho_j^H)^h \nu_{j,t}^H. \quad (16)$$

Holding fundamentals and institutional absorption capacity locally constant, the initial valuation effect of the innovation is

$$m_{j,t}^H = \text{PIM}_{j,t-1} \nu_{j,t}^H = \text{IPI}_{j,t}^H. \quad (17)$$

Its expected remaining effect after h quarters is

$$E_t[m_{j,t+h}^H \mid \nu_{j,t}^H] = (\rho_j^H)^h \text{IPI}_{j,t}^H. \quad (18)$$

Let $R_{j,t \rightarrow t+h}^*$ denote the cumulative return attributable to changes in fundamental value. The expected cumulative return associated with the correction of the household-demand-induced valuation wedge is then

$$E_t[R_{j,t \rightarrow t+h} - R_{j,t \rightarrow t+h}^* \mid \nu_{j,t}^H] = - \left[1 - (\rho_j^H)^h \right] \text{IPI}_{j,t}^H. \quad (19)$$

Equation (19) provides the central testable implication. A positive household demand shock initially raises the stock price. As household demand subsequently mean reverts, the valuation effect dissipates and the stock earns lower future returns. The Price Impact Mul-

tiplier determines the magnitude of the initial distortion, while ρ_j^H determines how quickly that distortion reverses. Greater persistence delays the correction and shifts more of the reversal toward longer horizons.

The framework distinguishes the price effect of demand from its information content. If a demand shock reveals information about future cash flows, $p_{j,t}^*$ may move in the same direction as the observed price, reducing or eliminating the subsequent reversal. Thus, a positive contemporaneous price effect followed by little future underperformance is consistent with information-based demand. In contrast, a positive initial price effect followed by lower subsequent returns is consistent with a temporary valuation distortion.

2.5 Empirical Predictions

The framework produces two central predictions.

Prediction 1: Contemporaneous price impact. Holding the magnitude of the household demand shock fixed, its contemporaneous price effect is increasing in the Price Impact Multiplier:

$$\frac{\partial \Delta p_{j,t}}{\partial \nu_{j,t}^H} = \text{PIM}_{j,t-1}. \quad (20)$$

Household shocks should therefore produce larger price increases when institutional absorption capacity is limited.

Prediction 2: Subsequent return reversal. Household-demand-implied price impact should predict lower subsequent returns:

$$\frac{\partial E_t[R_{j,t \rightarrow t+h} - R_{j,t \rightarrow t+h}^*]}{\partial \nu_{j,t}^H} = - \left[1 - (\rho_j^H)^h \right] \text{PIM}_{j,t-1}. \quad (21)$$

The magnitude of the eventual reversal increases with the Price Impact Multiplier, while demand persistence governs the timing of the correction. These predictions motivate the empirical analysis of contemporaneous price effects, horizon-specific future returns, owner-

ship dynamics, and risk-adjusted portfolio performance.

3 Data

This section describes the data sources and variable construction. I classify the data into three categories: stock prices and firm fundamentals, institutional holdings, and analyst expectations. Table 1 summarizes the coverage and key dimensions of each dataset.

3.1 Prices and Fundamentals

Stock returns, prices, shares outstanding, and market capitalization are obtained from the Center for Research in Security Prices (CRSP). Accounting information is obtained from the Compustat North America Fundamentals Annual and Quarterly databases. To avoid look-ahead bias, accounting variables are matched to returns using the most recently available public information.

The sample is restricted to ordinary common shares listed on the NYSE, AMEX, and Nasdaq. Firms with missing return, price, or required accounting information are excluded from the relevant analysis. The underlying CRSP–Compustat universe spans 1980Q1 through 2024Q3. Tests that combine these data with the demand-system and analyst data use the corresponding common sample, which ends in 2017Q4.

Firm characteristics are selected to match the demand-system asset-pricing literature. The baseline characteristic set includes market beta, firm size, profitability, investment, dividends relative to book equity, and momentum. These variables capture dimensions of firm fundamentals and risk that may influence portfolio demand.

The empirical analysis examines both contemporaneous price adjustment and future stock returns. For horizon h , cumulative future returns are defined as

$$R_{j,t \rightarrow t+h} = \prod_{k=1}^h (1 + r_{j,t+k}) - 1, \quad (22)$$

where $r_{j,t+k}$ is the realized return of firm j in quarter $t+k$. Thus, the one-quarter horizon is the next-quarter return, whereas the twenty-quarter horizon is the five-year buy-and-hold return. I also examine quarter-by-quarter returns, $r_{j,t+h}$, to identify the timing of return predictability without aggregating returns across intermediate quarters.

Panel A of Table 1 summarizes the stock-level universe. It contains 714,468 firm-quarter observations representing 19,161 unique firms. The average quarter contains approximately 3,991 firms. Firm size is highly dispersed: average market equity is \$3.35 billion, compared with a median of \$174 million.

3.2 Institutional Holdings

Institutional holdings are obtained from Thomson Reuters 13F filings. Institutional investment managers with more than \$100 million in reportable securities are required to submit Form 13F to the Securities and Exchange Commission. These filings report long equity positions and provide broad coverage of institutional ownership in U.S. equity markets.

Institutional holdings are merged with the CRSP–Compustat universe using security identifiers. For each institution-quarter, I calculate the market value of every position and aggregate across positions to obtain the institution’s total assets under management (AUM). Portfolio weights are defined as the fraction of the institution’s equity portfolio invested in a given stock.

Institutional investors are classified into mutual funds, banks, insurance companies, pension funds, investment advisors, and other institutions. These classifications are used to examine heterogeneity in latent demand and estimated demand elasticity across investor groups.

For each stock-quarter, I measure household holdings as the difference between total shares outstanding and aggregate institutional holdings reported in the 13F data. This residual does not identify retail ownership perfectly because it also includes institutions that are not required to file Form 13F. Accordingly, the measure should be interpreted as

the residual non-13F sector. For expositional simplicity, I refer to it as household holdings throughout the paper.

Panel B of Table 1 summarizes the holdings data. The sample contains 13,417 unique institutions, 458,624 institution-quarter portfolios, and more than 93 million institution-stock-quarter observations. The average institution holds approximately 203 stocks per quarter, whereas the median holds 75. Institutional size is similarly dispersed: average AUM is approximately \$4.08 billion, compared with a median of \$258 million.

3.3 Analyst Coverage

Analyst-forecast data are obtained from the LSEG I/B/E/S Academic Detail History database through WRDS. For each firm-quarter, analyst coverage is defined as the number of distinct analysts issuing one-year-ahead earnings forecasts during the calendar quarter. I/B/E/S forecasts are linked to the CRSP-Compustat universe using the eight-character CUSIP. After completing the CRSP firm-quarter panel, observations without an I/B/E/S forecast are assigned zero analyst coverage.

Analyst coverage measures the firm’s information environment and the extent of professional monitoring. Firms followed by more analysts generally receive greater information production and scrutiny, whereas firms with little or no coverage operate in weaker informational environments. I therefore use analyst coverage as one conventional proxy for limits to arbitrage.

A firm is classified as having low analyst coverage when its coverage is less than or equal to the within-quarter median among firms with positive coverage. Zero-coverage firm-quarters are retained and assigned to the low-coverage group. Panel C of Table 1 shows that the matched analyst panel contains 613,022 firm-quarter observations and 17,268 unique firms. Mean analyst coverage is 3.83 analysts, the median is one analyst, and approximately 61.6% of firm-quarters have positive coverage.

Table 1: Data Sources and Sample Overview

Statistic	Description	Value
<i>Panel A: CRSP–Compustat stock universe</i>		
Sample period	Calendar-quarter coverage	1980Q1–2024Q3
Number of quarters	Quarters in the underlying stock universe	179
Firm-quarter observations	Stock-quarter observations with required data	714,468
Unique firms	Distinct CRSP firms	19,161
Average firms per quarter	Mean cross-sectional firm count	3,991
Median firms per quarter	Median cross-sectional firm count	3,938
Mean market equity	Firm market capitalization, \$ millions	3,350
10th percentile market equity	Firm market capitalization, \$ millions	10
Median market equity	Firm market capitalization, \$ millions	174
90th percentile market equity	Firm market capitalization, \$ millions	4,235
<i>Panel B: Thomson Reuters 13F institutional holdings</i>		
Unique institutions	Distinct institutional investment managers	13,417
Institution-quarter portfolios	Institution-quarter portfolio observations	458,624
Institution–stock–quarter observations	Individual reported stock positions	>93 million
Mean stocks per portfolio	Average number of stocks held per institution-quarter	203
Median stocks per portfolio	Median number of stocks held per institution-quarter	75
Mean institutional AUM	Equity assets under management, \$ millions	4,076
Median institutional AUM	Equity assets under management, \$ millions	258
<i>Panel C: LSEG I/B/E/S analyst coverage</i>		
Sample period	Coverage in the matched demand-system panel	1980Q1–2017Q4
Number of quarters	Quarters represented in the analyst panel	152
Firm-quarter observations	CRSP firm-quarters after completing zero coverage	613,022
Unique firms	Distinct CRSP firms in the analyst panel	17,268
Mean analyst coverage	Distinct one-year-ahead analysts per firm-quarter	3.83
Median analyst coverage	Distinct one-year-ahead analysts per firm-quarter	1
Positive-coverage observations	Percentage of firm-quarters with at least one analyst	61.6%

This table summarizes the principal datasets used in the analysis. Panel A reports the underlying CRSP–Compustat stock universe. Market equity is measured in millions of dollars. The common sample used in individual regressions may be smaller because of lag requirements and the availability of demand, return, fundamental, or analyst variables.

Panel B summarizes institutional portfolios constructed from Thomson Reuters 13F filings. AUM is the aggregate market value of an institution’s reported equity positions and is measured in millions of dollars. Household holdings are defined as shares outstanding minus aggregate reported 13F holdings and therefore represent the residual non-13F sector rather than a pure measure of retail ownership.

Panel C summarizes the analyst-coverage panel constructed from LSEG I/B/E/S Detail History. Coverage is the number of distinct analysts issuing one-year-ahead earnings forecasts during the calendar quarter. Firm-quarters without an I/B/E/S forecast are assigned zero coverage after linkage to the CRSP–Compustat universe.

4 Empirical Results

This section presents the paper’s main empirical results. The central objective is to determine whether institutional absorption capacity amplifies the price effects of otherwise comparable household demand shocks. I begin by describing the estimation of investor-level demand systems, excess demand, and demand elasticities. I then examine the contemporaneous and future-return consequences of aggregate, household, and institutional excess demand. Finally, I construct the Price Impact Multiplier and test whether household-demand-implied price impact explains cross-sectional variation in equity mispricing, as predicted by equations (11), (13), and (19).

4.1 Demand-System Estimation

The first step is to estimate investors’ excess demand and demand elasticities. Following Kojen and Yogo (2019), I estimate the following demand system separately for each investor and quarter:

$$\log\left(\frac{w_{i,j,t}}{w_{i,0,t}}\right) = \alpha_{i,t} + b_{i,t}^{ME} \log(ME_{j,t}) + \beta'_{i,t} \mathbf{X}_{j,t} + d_{i,j,t}, \quad (23)$$

where $w_{i,j,t}$ is investor i ’s portfolio weight in stock j , $w_{i,0,t}$ is the investor’s portfolio weight in the outside asset, and $ME_{j,t}$ is the market equity of stock j . The vector $\mathbf{X}_{j,t}$ contains observed firm characteristics, including beta, profitability, investment, and dividends relative to book equity. Because market equity contains the stock price and is simultaneously determined with investor demand, I use instrumented market equity in estimating equation (23).

The demand system decomposes an investor’s demand for a stock into three components. The first is demand associated with observed firm characteristics, captured by $\beta'_{i,t} \mathbf{X}_{j,t}$. The coefficients are investor-specific, allowing investors to differ in their preferences for firm characteristics. The second component is the response of demand to price, identified by $b_{i,t}^{ME}$. Because portfolio weights incorporate the value of the position, the approximate elasticity

magnitude of shares demanded is

$$\eta_{i,t} \simeq 1 - b_{i,t}^{ME}. \quad (24)$$

A larger $\eta_{i,t}$ indicates that investor i adjusts the number of shares demanded more strongly in response to changes in price.

The residual, $d_{i,j,t}$, is the component of investor i 's demand for stock j that is not explained by instrumented prices or observed firm characteristics. I refer to this residual as *excess demand*. A positive value indicates that the investor allocates more of the portfolio to the stock than the demand system predicts, whereas a negative value indicates an allocation below the model-implied benchmark. Excess demand may reflect private information, idiosyncratic preferences, sentiment, or other unobserved determinants of portfolio choice. It should not, by itself, be interpreted as irrational demand.

Firm-level aggregate excess demand is the assets-under-management-weighted average of investor-level excess demand:

$$AED_{j,t} = \sum_{i \in \mathcal{I}_{j,t}} \omega_{i,j,t} d_{i,j,t}, \quad \omega_{i,j,t} = \frac{AUM_{i,t}}{\sum_{\ell \in \mathcal{I}_{j,t}} AUM_{\ell,t}}, \quad (25)$$

where $\mathcal{I}_{j,t}$ denotes the set of investors holding stock j in quarter t . This weighting scheme gives greater importance to demand from investors with larger portfolios.

I construct household and institutional excess demand analogously. Household excess demand, $HED_{j,t}$, is obtained from the residual non-13F household sector, while institutional excess demand, $IED_{j,t}$, aggregates the estimated residuals of 13F institutions. These sector-level measures identify which side of the market generates the excess-demand state observed for a stock.

Excess demand is conceptually different from trading flow. Trading measures a change in observed ownership between two dates, whereas excess demand measures the level of desired ownership relative to the demand-system benchmark. An investor can therefore have positive excess demand while reducing ownership if the remaining position continues

to exceed the model-implied allocation. Conversely, a purchase need not represent positive excess demand if it is fully explained by a change in price or firm characteristics. For this reason, I subsequently isolate innovations in household excess demand to identify newly arriving household demand more closely.

Let the household excess-demand process be

$$HED_{j,t} = \alpha_t + \rho_t^H HED_{j,t-1} + \nu_{j,t}^H, \quad (26)$$

where $\nu_{j,t}^H$ is the household demand innovation. The innovation captures the component of current household excess demand that is not predicted by its lagged value. To express this innovation as a quantity relative to stock supply, I scale it by lagged household ownership:

$$\nu_{j,t}^{H,q} = s_{j,t-1}^H \nu_{j,t}^H, \quad (27)$$

where $s_{j,t-1}^H$ is the household share of the stock at the end of the previous quarter.

On the institutional side, I combine investor-level elasticities with institutional ownership to construct institutional absorption capacity:

$$\kappa_{j,t-1}^I = \sum_{i \in \mathcal{I}_{j,t-1}^I} s_{i,j,t-1} \eta_{i,t-1}, \quad (28)$$

where $s_{i,j,t-1}$ is institution i 's ownership as a fraction of shares outstanding. Thus, absorption capacity is high when institutions own a large fraction of the stock and their demand is relatively responsive to price.

The Price Impact Multiplier is the inverse of institutional absorption capacity. In the empirical implementation, I use the regularized measure

$$\text{PIM}_{j,t-1} = \left[\max \left\{ \kappa_{j,t-1}^I, Q_{0.05,t-1}(\kappa^I) \right\} \right]^{-1}. \quad (29)$$

The quarter-specific fifth-percentile floor limits the influence of mechanically extreme values generated by near-zero estimated capacity. The household-demand-implied price impact is then

$$\text{IPI}_{j,t}^H = \text{PIM}_{j,t-1} \times \nu_{j,t}^{H,q}. \quad (30)$$

This measure is high when newly arriving household demand is large relative to the elastic institutional capital available to absorb it.

4.2 Summary Statistics

Table 2 summarizes the principal variables. For each quarter, I first compute the indicated cross-sectional statistic across firms and then average that statistic over the sample period. The table therefore describes the cross-sectional distribution observed in a typical quarter rather than the pooled time-series distribution.

Panel A reports aggregate, household, and institutional excess demand. The sample contains approximately 4,000 firms per quarter over 152 quarters. Aggregate excess demand has a mean of -0.01 and a standard deviation of 0.49 , indicating substantial within-quarter variation. Its average cross-sectional median is slightly positive, whereas its mean is negative, suggesting a left-skewed distribution. The average cross-sectional minimum and maximum are -4.66 and 2.30 , respectively. Household and institutional excess demand also exhibit substantial cross-sectional dispersion, providing the variation needed to examine their separate pricing consequences.

Panel B summarizes the estimated demand elasticities. Across investor-quarter estimates, the mean elasticity is 0.18 , the median is 0.02 , and the standard deviation is 0.27 . The difference between the mean and median reflects a highly skewed distribution in which many estimated elasticities are close to the lower estimation boundary. Consequently, the simple median provides an incomplete description of the economically relevant variation in elastic institutional capital.

The aggregate institutional sector has an average elasticity of 0.09, compared with 0.58 for the household residual sector. Thus, household demand is approximately six times as responsive to price as institutional demand. Economically, a 1% increase in price is associated with an estimated 0.58% reduction in household shares demanded but only a 0.09% reduction in institutional shares demanded. Although institutions' large ownership shares contribute to their total absorption capacity, their low estimated elasticity limits how strongly their demand responds to a household shock at the prevailing price.

Panel B also reports the ownership-weighted elasticity of the investors holding each stock. The average firm-level elasticity is 0.35, with a cross-sectional standard deviation of 0.15, and ranges from an average minimum of 0.06 to an average maximum of 0.64. This dispersion implies that otherwise similar household shocks can face substantially different demand responses from the investors holding the affected stocks. The table also reports institutional absorption capacity and the corresponding Price Impact Multiplier, which translate this elasticity variation into the central state variable used in the subsequent analysis.

Panel C reports the firm characteristics used as controls throughout the empirical analysis: profitability, investment, dividend payout, log market equity, momentum, analyst coverage, institutional ownership, and idiosyncratic volatility. The substantial cross-sectional variation in these characteristics is also useful for distinguishing institutional absorption capacity from conventional limits-to-arbitrage proxies.

Finally, Panel D summarizes future stock returns over horizons ranging from one quarter to five years. Both average returns and cross-sectional dispersion increase with the return horizon. The sample continues to contain a broad cross-section of firms at longer horizons, permitting an examination of the short- and long-run return consequences of household-demand-implied price impact.

Table 2: Summary Statistics for Excess Demand, Absorption Capacity, Implied Price Impact, Controls, and Returns

Variable	N_q	Avg. N	Mean	Std. Dev.	Min	Median	Max
<i>Panel A: Excess Demand</i>							
Aggregate excess demand (AED)	152	3,975	-0.01	0.49	-4.66	0.02	2.30
Household excess demand (HED)	152	3,975	-0.00	0.85	-8.63	0.07	2.88
Institutional excess demand (IED)	152	3,975	0.87	1.14	-5.44	0.90	7.91
<i>Panel B: Demand Elasticities and Absorption Capacity</i>							
Investor-level elasticity [†]	–	10,206	0.18	0.27	0.01	0.02	2.66
Household-sector elasticity	152	1	0.58	0.40	0.24	0.38	2.23
Institutional-sector elasticity	152	1	0.09	0.05	0.01	0.08	0.23
Stock-level ownership-weighted elasticity	152	3,969	0.35	0.15	0.06	0.33	0.64
Institutional absorption capacity (κ^I)	152	3,969	0.04	0.03	0.00	0.03	0.25
<i>Panel C: Price Impact Multiplier and Implied Price Impact</i>							
asinh(PIM)	149	3,836	4.78	1.38	3.06	4.33	8.35
asinh(IPI ^H)	149	3,836	0.22	2.95	-6.65	0.48	6.68
<i>Panel D: Controls</i>							
Profitability	152	3,975	0.11	0.37	-1.45	0.17	0.72
Asset growth	152	3,975	0.09	0.24	-0.48	0.07	0.82
Dividend payout	152	3,975	0.02	0.03	0.00	0.00	0.10
Log market equity	152	3,975	5.22	2.04	-0.65	5.09	12.16
Momentum	152	3,967	0.01	0.47	-3.15	0.05	2.29
Analyst coverage	152	3,975	3.86	5.16	0.00	1.83	36.76
Institutional ownership	152	3,969	0.39	0.27	0.00	0.38	1.29
Idiosyncratic volatility	152	3,975	0.03	0.02	0.00	0.03	0.33
<i>Panel E: Future Returns</i>							
Next-quarter return	152	3,892	0.04	0.27	-0.85	0.02	4.45
One-year cumulative return	152	3,643	0.16	0.64	-0.95	0.08	11.85
Three-year cumulative return	152	3,077	0.50	1.40	-0.99	0.26	29.72
Five-year cumulative return	152	2,633	0.95	2.36	-1.00	0.50	52.74

This table reports time-series averages of quarterly cross-sectional moments. For each variable and calendar quarter, I calculate the number of nonmissing firms and the cross-sectional mean, standard deviation, minimum, median, and maximum. The table then averages each statistic across quarters. Thus, the reported minimum and maximum are averages of quarterly extrema rather than full-sample extrema. N_q is the number of available quarters, and Avg. N is the average quarterly cross-sectional number of firms. For the investor-level elasticity row, denoted by [†], the Avg. N column instead reports the number of investor-level elasticity estimates.

Aggregate excess demand, $AED_{j,t}$, is the holdings-weighted firm-level excess-demand measure across investors. Household excess demand, $HED_{j,t}$, is the corresponding measure for the residual non-13F household sector, and institutional excess demand, $IED_{j,t}$, aggregates all 13F institutions. Excess demand is the component of portfolio demand not explained by instrumented stock prices or observed firm characteristics. Investor-level elasticity, $\eta_{i,t}$, is the estimated magnitude of the percentage change in shares demanded associated with a 1% change in price. Household- and institutional-sector elasticities are ownership-weighted sector averages. Stock-level elasticity is the ownership-weighted elasticity across all investors holding the stock. Institutional absorption capacity is

$$\kappa_{j,t}^I = \sum_{i \in \mathcal{I}_{j,t}^I} s_{i,j,t} \eta_{i,t},$$

where $s_{i,j,t}$ is institution i 's ownership as a fraction of shares outstanding. The regularized Price Impact Multiplier is

$$\text{PIM}_{j,t-1} = [\max \{ \kappa_{j,t-1}^I, Q_{0.05,t-1}(\kappa^I) \}]^{-1}.$$

The household-demand-implied price impact is

$$\text{IPI}_{j,t}^H = \text{PIM}_{j,t-1} \nu_{j,t}^H,$$

4.3 Excess Demand and the Term Structure of Stock Returns

I begin by documenting the relation between excess demand and stock returns before introducing the Price Impact Multiplier. This baseline analysis establishes whether excess demand is associated with contemporaneous price movements, whether those movements subsequently reverse, and whether the return patterns differ between households and institutions. Motivated by equations (6) and (19), I estimate the following horizon-specific Fama–MacBeth regressions:

$$r_{j,t+h} = a_{t,h} + \beta_h^g ED_{j,t}^g + \Gamma_h' X_{j,t}^{(h)} + u_{j,t+h}, \quad h \in \{0, 1, \dots, 12\}, \quad (31)$$

where $r_{j,t+h}$ is the raw return on stock j realized exactly h quarters after the excess-demand measurement. The specification with $h = 0$ examines the contemporaneous return association, while specifications with $h = 1, \dots, 12$ trace the quarter-by-quarter term structure of subsequent returns. The variable $ED_{j,t}^g$ denotes excess demand for investor sector g . I first consider aggregate excess demand, $AED_{j,t}$, and then separate it into household excess demand, $HED_{j,t}$, and institutional excess demand, $IED_{j,t}$. I also estimate the regressions separately for each institutional investor type.

The vector $X_{j,t}^{(h)}$ contains profitability, asset growth, dividend payout, log market equity, and momentum. In the contemporaneous specification, $X_{j,t}^{(0)} = X_{j,t-1}$, because characteristics measured during quarter t may be jointly determined with current returns and holdings. For future returns, $X_{j,t}^{(h)} = X_{j,t}$ for $h \geq 1$. Each excess-demand measure is standardized within quarter. Consequently, β_h^g is the percentage-point difference in the quarterly return associated with a one-standard-deviation increase in the indicated excess-demand measure. Table 3 reports the results.

Panel A first considers aggregate excess demand. A one-standard-deviation increase in aggregate excess demand is associated with a 5.00-percentage-point higher contemporaneous return, with a t -statistic of 16.19. This positive association is consistent with the market-

clearing prediction that greater excess demand is accompanied by a higher equilibrium price. Because excess demand and returns are measured during the same quarter, however, this coefficient is descriptive and should not be interpreted as a causal price-impact estimate.

Aggregate excess demand predicts significantly lower returns after the formation quarter. The next-quarter coefficient is -0.66 percentage points, implying that stocks with one-standard-deviation higher excess demand underperform by approximately 66 basis points in the following quarter. The coefficients remain negative and statistically significant through quarter 11, although their magnitudes do not decline monotonically. Summing the 12 quarter-specific coefficients yields approximately -4.53 percentage points, close to the 5.00-percentage-point contemporaneous association. This comparison is descriptive rather than a formal cumulative-return estimate because each coefficient comes from a separate horizon-specific regression. Nevertheless, it indicates that much of the positive contemporaneous return associated with aggregate excess demand is subsequently reversed.

The next two rows separate household and institutional excess demand. Household excess demand has the larger contemporaneous association: a one-standard-deviation increase is associated with a 3.95-percentage-point higher contemporaneous return, compared with 2.91 percentage points for institutional excess demand. Because the signals are standardized separately, these magnitudes compare the return effect of a typical within-quarter realization of each measure rather than an identical quantity shock.

Household excess demand also accounts for most of the subsequent reversal. Its next-quarter coefficient is -0.42 percentage points and remains negative over the following quarters. The response is non-monotonic: it initially declines in magnitude, becomes more negative again around quarters four through seven, and then gradually approaches zero. Household excess demand predicts significantly lower returns for approximately ten quarters. Thus, high household excess demand is associated with a sharp contemporaneous price increase followed by a gradual period of underperformance lasting nearly three years.

Institutional excess demand exhibits a different pattern. Although it is strongly positively

associated with contemporaneous returns, none of its future-return coefficients is statistically significant when institutional excess demand is examined separately. Institutional excess demand therefore does not generate the persistent return reversal observed for household excess demand. This pattern is consistent with institutional excess demand containing more information about fundamental value, although the return regressions alone cannot distinguish information-based demand from other explanations.

Panel B decomposes institutional excess demand by investor type. Investment advisors have the largest contemporaneous coefficient, at 1.17 percentage points, followed by mutual funds at 0.96 percentage points. The remaining institutional categories also have positive and statistically significant contemporaneous coefficients. Across investor types, however, there is little evidence of a stable future-return reversal. Most subsequent coefficients are statistically insignificant, and the isolated significant estimates do not form a systematic term structure. The absence of persistent underperformance is therefore a broad feature of institutional excess demand rather than a result driven by a single institutional category.

Panel C includes household and aggregate institutional excess demand jointly. The household coefficient becomes slightly larger in magnitude and remains statistically significant after controlling for institutional excess demand. A one-standard-deviation increase in household excess demand predicts a -0.47 -percentage-point next-quarter return and continues to predict lower returns at longer horizons. Conditional institutional excess demand predicts a -0.27 -percentage-point next-quarter return, but this association largely disappears after the first quarter. Controlling for common variation between the two sectors therefore reinforces the central distinction: household excess demand is associated with a persistent return reversal, whereas institutional excess demand is not.

These results establish the baseline return pattern but do not yet identify the paper's central mechanism. In particular, the regressions do not distinguish newly arriving household demand from an existing excess-demand state, nor do they account for differences in the institutional capital available to accommodate a household shock. The following subsec-

tions isolate innovations in household excess demand and use the Price Impact Multiplier to test whether otherwise comparable household shocks produce larger price distortions when institutional absorption capacity is limited.

Table 3: Contemporaneous and Future Return Associations of Aggregate, Household, and Institutional Excess Demand

Excess-demand measure	Return realized exactly h quarters after formation												
	t												
Panel A: Aggregate and sector-level excess demand													
Aggregate excess demand	5.00*** (16.19)	-0.66*** (-5.98)	-0.41*** (-3.65)	-0.46*** (-3.88)	-0.42*** (-3.45)	-0.49*** (-4.03)	-0.40*** (-3.35)	-0.37*** (-2.87)	-0.25*** (-2.03)	-0.30*** (-2.44)	-0.33*** (-2.51)	-0.25*** (-2.03)	-0.19 (-1.50)
Household excess demand	3.95*** (16.97)	-0.42*** (-4.18)	-0.27*** (-2.64)	-0.32*** (-2.95)	-0.29*** (-2.71)	-0.35*** (-3.28)	-0.30*** (-2.77)	-0.29*** (-2.48)	-0.18* (-1.70)	-0.22*** (-2.12)	-0.21* (-1.88)	-0.15 (-1.37)	-0.13 (-1.13)
Institutional excess demand	2.91*** (14.47)	-0.16 (-1.57)	-0.04 (-0.36)	-0.06 (-0.53)	-0.10 (-0.83)	-0.10 (-0.93)	-0.05 (-0.52)	-0.07 (-0.67)	-0.02 (-0.28)	-0.02 (-0.16)	-0.12 (-1.13)	-0.11 (-1.07)	-0.09 (-0.88)
Panel B: Institutional excess demand by investor type													
Investment advisors	1.17*** (9.72)	-0.11* (-1.75)	-0.02 (-0.27)	0.02 (0.37)	-0.04 (-0.60)	-0.10 (-1.51)	0.01 (0.19)	0.07 (1.12)	0.04 (0.65)	0.11 (1.65)	0.04 (0.50)	0.03 (0.37)	0.01 (0.10)
Mutual funds	0.96*** (5.26)	-0.13 (-1.30)	-0.01 (-0.10)	-0.03 (-0.31)	0.04 (0.41)	-0.02 (-0.21)	0.07 (0.71)	0.07 (0.73)	0.06 (0.56)	0.09 (0.91)	0.10 (0.98)	0.14 (1.34)	0.06 (0.57)
Pension funds	0.54*** (6.21)	0.05 (0.62)	0.09 (1.39)	0.04 (0.50)	0.02 (0.23)	0.01 (0.10)	0.05 (0.63)	0.02 (0.24)	0.04 (0.58)	-0.00 (-0.07)	0.02 (0.30)	0.02 (0.38)	0.04 (0.65)
Other institutions	0.50*** (4.09)	-0.18* (-1.72)	-0.05 (-0.45)	-0.09 (-0.79)	0.01 (0.06)	-0.04 (-0.37)	-0.14* (-1.93)	-0.19** (-2.14)	-0.10 (-1.11)	-0.03 (-0.39)	-0.07 (-0.54)	0.09 (0.83)	0.00 (0.01)
Banks	0.43*** (4.90)	-0.03 (-0.52)	0.03 (0.42)	0.05 (0.77)	-0.01 (-0.20)	-0.00 (-0.07)	0.04 (0.56)	0.06 (0.96)	0.05 (0.77)	0.02 (0.27)	-0.05 (-0.88)	-0.06 (-1.05)	-0.06 (-0.83)
Insurance companies	0.36*** (4.03)	-0.12* (-1.75)	0.05 (0.88)	0.08 (1.26)	0.13* (1.94)	0.13** (1.98)	0.02 (0.31)	0.05 (0.70)	-0.03 (-0.51)	0.01 (0.14)	0.03 (0.52)	-0.03 (-0.43)	-0.03 (-0.41)
Panel C: Joint household-institutional excess-demand horse race													
Household excess demand institutional excess demand	4.43*** (17.50)	-0.47*** (-4.79)	-0.29*** (-2.97)	-0.33*** (-3.33)	-0.33*** (-3.29)	-0.38*** (-3.87)	-0.33*** (-3.25)	-0.32*** (-2.89)	-0.21*** (-2.05)	-0.25*** (-2.52)	-0.24*** (-2.23)	-0.18* (-1.72)	-0.15 (-1.42)
Institutional excess demand household excess demand	3.60*** (17.32)	-0.27*** (-2.74)	-0.10 (-1.14)	-0.13 (-1.28)	-0.17 (-1.58)	-0.18* (-1.86)	-0.13 (-1.42)	-0.13 (-1.50)	-0.07 (-0.81)	-0.07 (-0.77)	-0.17* (-1.70)	-0.16 (-1.61)	-0.12 (-1.39)

Panels A and B report separate horizon-specific Fama-MacBeth regressions of stock returns on the indicated excess-demand measure. The first column reports the raw stock return realized during formation quarter t and uses controls measured at $t - 1$. It is a contemporaneous association rather than a predictive or causal estimate. Columns $h = 1, \dots, 12$ report the raw stock return realized exactly h quarters after formation and use controls measured in formation quarter t .

Excess-demand signals are standardized within quarter. Entries are therefore percentage-point return differences associated with a one-standard-deviation increase in the indicated measure. Controls are profitability, asset growth, dividend payout, log market equity, and momentum. Type-specific excess demand is treated as missing when no measure is observed for that investor type and is not set equal to zero. Consequently, sample coverage differs across the institutional-type rows.

Panel C includes household and aggregate institutional excess demand simultaneously. Aggregate excess demand is the holdings-weighted firm-level measure across investors. Household excess demand is measured for the residual non-13F household sector, while institutional excess demand aggregates the 13F institutional sector. Coefficients are time-series averages of quarterly cross-sectional slopes. Parentheses report Newey-West t -statistics with four lags. Significance levels are *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

4.4 Predictable and Innovative Components of Excess Demand

The preceding subsection shows that household and institutional excess demand are both associated with substantially higher contemporaneous returns, but their subsequent return patterns differ sharply. Household excess demand predicts lower returns for approximately three years, whereas institutional excess demand exhibits little systematic association with returns beyond the following quarter. I next examine whether this difference can be explained by the persistence of the two excess-demand processes.

Excess demand contains two conceptually distinct components. An innovation captures the newly arriving component of demand and should have its strongest price effect when it first reaches the market. The predictable component is inherited from the previous quarter's excess-demand state. It can remain elevated without additional buying and therefore captures the persistence of investors' relative preference for the stock. Separating these components distinguishes the immediate price response to newly arriving demand from the subsequent return consequences of an existing excess-demand state.

For households and institutions, I estimate the following quarter-specific projection:

$$ED_{j,t}^s = \alpha_t^s + \rho_t^s ED_{j,t-1}^s + \nu_{j,t}^s, \quad s \in \{H, I\}, \quad (32)$$

where $ED_{j,t}^H = HED_{j,t}$ denotes household excess demand and $ED_{j,t}^I = IED_{j,t}$ denotes aggregate institutional excess demand. The projection is estimated separately in each quarter using firms with consecutive observations. The time-series averages of the quarterly estimates are

$$\begin{aligned} HED_{j,t} &= -0.00 + 0.88 HED_{j,t-1} + \nu_{j,t}^H, & \overline{R^2} &= 0.75, \\ IED_{j,t} &= 0.15 + 0.84 IED_{j,t-1} + \nu_{j,t}^I, & \overline{R^2} &= 0.70. \end{aligned} \quad (33)$$

Both excess-demand processes are highly persistent. The average persistence coefficient is 0.88 for households and 0.84 for institutions, while the average cross-sectional R^2 is 75% and 70%, respectively. At the household persistence estimate, approximately 80% of an

excess-demand shock dissipates after about 12 quarters. Institutional excess demand decays somewhat faster but is also strongly persistent. Differences in persistence alone are therefore unlikely to explain why household excess demand predicts a prolonged return reversal while institutional excess demand does not.

For each sector, I define the predictable component as

$$\widehat{ED}_{j,t}^{s,P} = \hat{\rho}_t^s ED_{j,t-1}^s, \quad (34)$$

and the innovation as

$$\nu_{j,t}^s = ED_{j,t}^s - \hat{\alpha}_t^s - \hat{\rho}_t^s ED_{j,t-1}^s. \quad (35)$$

The quarter-specific intercept is absorbed by the intercept in the subsequent cross-sectional regressions. The predictable component captures the excess-demand state inherited from the previous quarter, whereas the innovation captures newly arriving excess demand during quarter t .

I include the two components jointly in horizon-specific Fama–MacBeth regressions:

$$r_{j,t+h} = a_{t,h} + \beta_h^{P,s} \widehat{ED}_{j,t}^{s,P} + \beta_h^{N,s} \nu_{j,t}^s + \Gamma_h' X_{j,t}^{(h)} + u_{j,t+h}, \quad h = 0, \dots, 12. \quad (36)$$

The predictable and innovation components are standardized independently within quarter. The coefficients therefore measure percentage-point return differences associated with a one-standard-deviation increase in one component, conditional on the other. As before, contemporaneous regressions use controls measured at $t - 1$, while future-return regressions use controls measured in formation quarter t .

Table 4 reports the results. The first two rows of Panel A decompose aggregate excess demand. Predictable aggregate excess demand is associated with a 0.57-percentage-point lower contemporaneous return. Its future-return coefficients are also negative, at approximately 40–45 basis points per quarter over the first five quarters, and remain statistically significant

through approximately quarter 10. Conditional on newly arriving demand, an elevated inherited excess-demand state is therefore associated with an ongoing return correction rather than an additional contemporaneous price increase.

Aggregate excess-demand innovations produce a sharply different contemporaneous response. A one-standard-deviation innovation is associated with a 9.83-percentage-point higher contemporaneous return. The next-quarter coefficient is -0.55 percentage points, while the later coefficients are negative but non-monotonic. The innovation therefore captures most of the immediate positive price response and part of the subsequent reversal, although its longer-horizon pattern is less stable than that of predictable excess demand.

The household decomposition closely resembles the aggregate results. Predictable household excess demand is associated with a 0.37-percentage-point lower contemporaneous return and significantly lower returns over most of the following six quarters. Conditional on the innovation, a high inherited household excess-demand state appears to be in the correction phase: household demand remains above its model-implied benchmark, but the state no longer represents newly arriving demand.

Household excess-demand innovations, in contrast, are associated with an 8.27-percentage-point higher contemporaneous return. The next-quarter coefficient is -0.38 percentage points, with additional negative coefficients appearing in quarters five through seven and at several later horizons. The combination of a large contemporaneous increase and negative subsequent returns is consistent with newly arriving household demand producing an initial valuation effect that subsequently corrects.

The institutional decomposition is materially different. Predictable institutional excess demand has no significant contemporaneous or future-return association. Institutional excess-demand innovations are associated with a 4.78-percentage-point higher contemporaneous return and a 0.22-percentage-point lower next-quarter return, but the remaining future-return coefficients are generally small and statistically insignificant. Newly arriving institutional excess demand is therefore associated with an immediate price adjustment and

a modest short-term reversal, but not with the prolonged return pattern observed for household excess demand.

Panel B shows that this conclusion is broadly similar across institutional types. Innovations for every institutional category are positively associated with contemporaneous returns, but there is no stable longer-horizon reversal. Predictable institutional excess demand is also generally unrelated to future returns. The household–institutional difference therefore does not arise because one sector’s excess demand is persistent while the other’s is transitory: both sector-level processes are highly persistent.

The decomposition instead points to the initial pricing of newly arriving household demand. Household innovations have a substantially larger contemporaneous return association than institutional innovations and are followed by more persistent underperformance. The framework predicts that the magnitude of this initial valuation effect depends on institutional absorption capacity. The next subsection tests this mechanism by multiplying household excess-demand innovations by the Price Impact Multiplier.

4.5 Household Demand Shocks and the Price Impact Multiplier

The preceding results show that innovations in household excess demand are associated with a sharp contemporaneous price increase followed by lower subsequent returns. I now examine why otherwise similar household demand shocks produce different price effects across stocks. The conceptual framework predicts that the price response depends on both the size of the shock and the amount of elastic institutional capital available to absorb it. When the institutions holding a stock have relatively elastic demand, they can accommodate household trading with a limited price adjustment. When institutional demand is inelastic, a larger price movement is required to clear the market.

I begin with the quarter-specific decomposition

$$HED_{j,t} = \hat{\alpha}_t^H + \hat{\rho}_t^H HED_{j,t-1} + \nu_{j,t}^H, \quad (37)$$

Table 4: Contemporaneous and Future Return Associations of Predictable and Innovative Excess Demand

Excess-demand measure		Contemporaneous		Future return horizon h											
Component	t	1	2	3	4	5	6	7	8	9	10	11	12		
Panel A: Aggregate and sector-level excess demand															
Aggregate excess demand	Predictable	-0.57*** (-4.91)	-0.42*** (-3.68)	-0.45*** (-3.80)	-0.44*** (-3.83)	-0.39*** (-3.34)	-0.30*** (-2.41)	-0.25*** (-1.99)	-0.30*** (-2.36)	-0.26* (-1.97)	-0.22* (-1.79)	-0.19 (-1.50)	-0.19 (-1.53)		
	Innovation	9.83*** (21.26)	-0.55*** (-7.78)	-0.14* (-1.80)	0.01 (0.08)	-0.30*** (-5.15)	-0.27*** (-3.83)	-0.29*** (-4.59)	0.03 (0.49)	-0.13* (-1.86)	-0.19*** (-3.36)	-0.17** (-2.48)	0.00 (0.00)		
Household excess demand	Predictable	-0.37*** (-3.51)	-0.24** (-2.40)	-0.32*** (-2.97)	-0.31*** (-3.03)	-0.28*** (-2.66)	-0.22* (-1.94)	-0.18 (-1.60)	-0.23** (-2.06)	-0.17 (-1.54)	-0.12 (-1.14)	-0.11 (-1.03)	-0.14 (-1.21)		
	Innovation	8.27*** (17.40)	-0.38*** (-5.60)	-0.09 (-1.29)	-0.02 (0.39)	-0.23*** (-4.91)	-0.20*** (-3.03)	-0.26*** (-4.73)	0.03 (0.62)	-0.11* (-1.96)	-0.13*** (-2.71)	-0.09 (-1.51)	-0.00 (-0.03)		
Institutional excess demand	Predictable	0.04 (0.39)	-0.08 (-0.79)	-0.09 (-0.75)	-0.11 (-1.05)	-0.07 (-0.72)	-0.05 (-0.47)	-0.03 (-0.29)	0.01 (0.13)	-0.08 (-0.85)	-0.13 (-1.20)	-0.08 (-0.83)	-0.05 (-0.58)		
	Innovation	4.78*** (16.23)	-0.22*** (-3.48)	0.09 (1.62)	-0.03 (-0.50)	-0.11* (-1.77)	-0.06 (-1.14)	-0.06 (-1.22)	-0.06 (-1.04)	0.01 (0.13)	-0.05 (-0.74)	-0.08 (-1.26)	-0.11 (-1.53)		
Panel B: Institutional excess demand by investor type															
Mutual funds	Predictable	-0.09 (-1.01)	-0.04 (-0.42)	-0.06 (-0.62)	0.01 (0.14)	-0.03 (-0.33)	0.05 (0.54)	0.08 (0.88)	0.09 (0.88)	0.05 (0.52)	0.10 (0.95)	0.16 (1.53)	0.03 (0.26)		
	Innovation	2.12*** (7.08)	-0.17*** (-2.79)	-0.00 (-0.04)	-0.05 (-0.55)	0.02 (0.39)	-0.11* (-1.71)	0.04 (0.62)	-0.09 (-1.53)	0.02 (0.38)	-0.03 (-0.47)	-0.10 (-1.15)	0.15* (1.92)		
Investment advisors	Predictable	-0.15** (-2.44)	-0.04 (-0.75)	-0.01 (-0.09)	-0.07 (-1.28)	-0.10 (-1.56)	-0.01 (-0.10)	0.04 (0.59)	0.04 (0.54)	0.13* (1.93)	0.05 (0.62)	0.05 (0.58)	0.04 (0.59)		
	Innovation	2.05*** (12.98)	-0.12** (-2.40)	0.00 (0.05)	0.09 (1.25)	-0.01 (-0.11)	-0.13** (-2.09)	-0.05 (-0.72)	0.05 (0.82)	-0.03 (-0.39)	0.04 (0.64)	-0.04 (-0.57)	0.01 (0.12)		
Pension funds	Predictable	-0.01 (-0.17)	0.06 (0.96)	0.02 (0.27)	0.03 (0.40)	0.01 (0.06)	0.00 (0.05)	0.01 (0.16)	0.04 (0.71)	0.02 (0.28)	0.05 (0.71)	0.03 (0.59)	0.05 (0.70)		
	Innovation	1.37*** (7.69)	-0.02 (-0.30)	0.05 (0.95)	-0.01 (-0.51)	-0.03 (-0.51)	0.02 (0.37)	-0.03 (-0.35)	-0.11 (-0.91)	0.00 (0.04)	-0.08 (-1.26)	0.01 (0.12)	0.01 (0.09)		
Insurance companies	Predictable	-0.22*** (-3.20)	0.03 (0.47)	0.06 (1.13)	0.13* (1.97)	0.12* (1.85)	0.07 (0.99)	0.06 (0.87)	-0.04 (-0.63)	0.03 (0.05)	0.03 (0.53)	-0.00 (-0.07)	-0.08 (-0.99)		
	Innovation	1.21*** (7.82)	-0.21*** (-3.33)	-0.01 (-0.17)	-0.03 (-0.48)	0.00 (0.02)	0.11* (1.85)	-0.06 (-1.02)	0.09 (1.15)	-0.02 (-0.40)	-0.14*** (-2.86)	0.04 (0.64)	-0.03 (-0.48)		
Banks	Predictable	-0.07 (-0.96)	0.02 (0.35)	0.02 (0.27)	-0.01 (-0.11)	0.00 (0.01)	0.03 (0.48)	0.08 (1.20)	0.05 (0.68)	0.01 (0.20)	-0.04 (-0.69)	-0.03 (-0.58)	-0.06 (-1.15)		
	Innovation	1.01*** (6.90)	-0.15*** (-2.88)	-0.01 (-0.15)	0.03 (0.57)	-0.02 (-0.39)	-0.05 (-0.87)	0.00 (0.10)	-0.01 (-0.16)	0.04 (0.87)	0.06 (1.16)	-0.05 (-1.09)	-0.01 (-0.12)		
Other institutions	Predictable	-0.18** (-2.00)	-0.17** (-2.15)	-0.15* (-1.83)	-0.05 (-0.42)	-0.04 (-0.37)	-0.05 (-0.67)	-0.13 (-1.45)	-0.16* (-1.80)	-0.12 (-1.27)	-0.08 (-0.91)	0.13 (1.09)	0.00 (0.04)		
	Innovation	0.58*** (4.02)	-0.08 (-1.01)	0.02 (0.18)	-0.04 (-0.41)	0.07 (0.71)	0.09 (1.01)	-0.12 (-1.37)	-0.09 (-1.36)	0.05 (0.65)	0.03 (0.35)	-0.08 (-0.84)	0.10 (0.94)		

For each excess-demand measure, I estimate the real-time, quarter-specific projection

$$ED_{j,t} = \hat{\alpha}_t + \hat{\rho}_t ED_{j,t-1} + \nu_{j,t}.$$

The predictable component is $\hat{\rho}_t ED_{j,t-1}$, while the innovation is the regression residual $\nu_{j,t}$. The quarter-specific intercept is absorbed by the intercept in the return regression. Both components are standardized independently within formation quarter. Entries therefore report percentage-point return differences associated with a one-standard-deviation increase in the indicated component, conditional on the other component.

The t column reports a Fama–MacBeth regression of the raw stock return realized during the formation quarter on the predictable and innovation components jointly. Controls in this contemporaneous specification are lagged one quarter. This column reports an association rather than a predictive or causal estimate. Columns $h = 1, \dots, 12$ report separate Fama–MacBeth regressions of the raw stock return realized exactly h quarters after formation; these specifications use controls measured in formation quarter t .

Controls are profitability, asset growth, dividend payout, log market equity, and momentum. Entries are time-series averages of quarterly cross-sectional slopes. Parentheses report Newey–West t -statistics with four lags. Aggregate and sector-level measures use 151 formation quarters. Type-specific samples vary because investor-type excess demand is observed only when that type holds the firm. Aggregate excess demand is the holdings-weighted all-investor measure; household excess demand is measured for the residual non-13F household sector; and institutional excess demand aggregates all 13F institutions. Significance levels are *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

where $\nu_{j,t}^H$ is the innovation in household excess demand and

$$\widehat{HED}_{j,t}^P = \hat{\rho}_t^H HED_{j,t-1} \quad (38)$$

is its predictable component. The innovation captures newly arriving household demand, whereas the predictable component captures the portion of the current excess-demand state inherited from the previous quarter.

Institutional absorption capacity is

$$\kappa_{j,t-1}^I = \sum_{i \in \mathcal{I}_{j,t-1}^I} s_{i,j,t-1} \eta_{i,t-1}, \quad (39)$$

where $s_{i,j,t-1}$ is institution i 's lagged ownership of stock j as a fraction of shares outstanding and $\eta_{i,t-1}$ is the magnitude of its estimated demand elasticity. Capacity is therefore high when a substantial fraction of the stock is held by institutions with price-elastic demand.

To prevent near-zero estimated capacity from mechanically producing extreme inverse-capacity values, I define regularized capacity as

$$\tilde{\kappa}_{j,t-1}^I = \max \left\{ \kappa_{j,t-1}^I, Q_{0.05,t-1}^+(\kappa^I) \right\}, \quad (40)$$

where $Q_{0.05,t-1}^+(\kappa^I)$ is the fifth percentile of positive institutional absorption capacity in quarter $t - 1$. The corresponding Price Impact Multiplier is

$$\text{PIM}_{j,t-1} = \left(\tilde{\kappa}_{j,t-1}^I \right)^{-1}. \quad (41)$$

Because $\nu_{j,t}^H$ is expressed in household demand-system units, I convert it into a quantity shock by multiplying it by lagged household ownership:

$$\nu_{j,t}^{H,q} = s_{j,t-1}^H \nu_{j,t}^H. \quad (42)$$

The household-demand-implied price impact is then

$$\text{IPI}_{j,t}^H = \text{PIM}_{j,t-1} \nu_{j,t}^{H,q} = \frac{s_{j,t-1}^H \nu_{j,t}^H}{\tilde{\kappa}_{j,t-1}^I}. \quad (43)$$

This measure is large when newly arriving household demand is large relative to the institutional capital available to absorb it.

To distinguish the role of absorption capacity from the magnitude of the underlying household innovation, I estimate

$$\begin{aligned} r_{j,t+h} = & a_{t,h} + \beta_h^\nu \nu_{j,t}^H + \beta_h^{IPI} \text{IPI}_{j,t}^H + \lambda_h M_{j,t-1}^H \\ & + \phi_h \widehat{HED}_{j,t}^P + \Gamma_h' X_{j,t}^{(h)} + u_{j,t+h}, \end{aligned} \quad (44)$$

where

$$M_{j,t-1}^H \equiv s_{j,t-1}^H \text{PIM}_{j,t-1} = \frac{s_{j,t-1}^H}{\tilde{\kappa}_{j,t-1}^I} \quad (45)$$

is the ownership-adjusted multiplier. Including $\nu_{j,t}^H$, $M_{j,t-1}^H$, and their product, $\text{IPI}_{j,t}^H$, produces the standard interaction specification. The coefficient β_h^{IPI} therefore tests whether the return response to household demand innovations varies systematically with institutional absorption capacity. The predictable household-demand component is included in the final specification to distinguish the incremental effect of newly arriving demand from the return predictability associated with the inherited demand state.

Table 5 provides three main results. First, newly arriving household demand is strongly associated with contemporaneous price increases. In the innovation-only specification, a one-standard-deviation household demand innovation is associated with an 11.21-percentage-point higher contemporaneous return. The implied-price-impact measure is also strongly positive when estimated separately, with a coefficient of 9.90 percentage points.

Second, the Price Impact Multiplier identifies substantial heterogeneity in the pricing of otherwise similar household innovations. When the innovation and implied price impact enter jointly, their contemporaneous coefficients are 8.86 and 6.44 percentage points, respectively.

Table 5: Household Demand Innovations, the Price Impact Multiplier, and Returns

	Contemporaneous return			Next-quarter return			
	Innovation only	Implied impact only	Joint	Innovation only	Implied impact only	Joint	Joint plus predictable demand
Household demand innovation	11.21*** (19.30)	–	8.86*** (17.04)	-0.47*** (-5.68)	–	-0.21*** (-2.96)	-0.17*** (-2.63)
Household-demand-implied price impact	–	9.90*** (22.39)	6.44*** (16.64)	–	-0.72*** (-8.06)	-0.64*** (-7.81)	-0.66*** (-7.96)
Predictable household excess demand	–	–	–	–	–	–	-0.23** (-2.14)
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Multiplier main effect	No	Yes	Yes	No	Yes	Yes	Yes
Average adjusted R^2	0.225	0.179	0.276	0.044	0.047	0.048	0.049
Quarters	149	149	149	149	149	149	149
Average cross-sectional N	3,829	3,829	3,829	3,752	3,752	3,752	3,752

This table reports Fama–MacBeth regressions of contemporaneous and next-quarter raw stock returns on household excess-demand innovations and household-demand-implied price impact. The household demand innovation is the residual $\nu_{j,t}^H$ from equation (37). Household-demand-implied price impact is

$$\text{IPI}_{j,t}^H = s_{j,t-1}^H \nu_{j,t}^H \text{PIM}_{j,t-1}.$$

The innovation-only specification includes the household demand innovation. The implied-impact-only specification includes $\text{IPI}_{j,t}^H$ and the multiplier main effect $M_{j,t-1}^H$. The joint specification includes the innovation, implied price impact, and multiplier main effect simultaneously. The final column additionally controls for predictable household excess demand, $\widehat{HED}_{j,t}^P$.

Institutional absorption capacity is measured at $t - 1$ and floored at its quarter-specific positive fifth percentile before the Price Impact Multiplier is constructed. Implied price impact is winsorized at the first and 99th percentiles within quarter. All reported continuous signals are standardized within quarter. Coefficients therefore report percentage-point return differences associated with a one-standard-deviation increase in the corresponding variable.

Contemporaneous regressions use profitability, asset growth, dividend payout, log market equity, and momentum measured at $t - 1$. Next-quarter regressions use the same controls measured in formation quarter t . Parentheses report Newey–West t -statistics with four lags. Median variance inflation factors for the household innovation and implied-price-impact variables are below 1.6 in the joint specifications. Significance levels are *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

The positive coefficient on implied price impact indicates that, conditional on the underlying innovation and the multiplier main effect, household shocks have larger contemporaneous return associations when institutional absorption capacity is limited. The joint specification also increases average adjusted R^2 from 0.225 in the innovation-only model to 0.276.

Third, the initial price increase is followed by a significant short-term reversal. Household-demand-implied price impact predicts a 0.72-percentage-point lower next-quarter return when estimated separately. In the joint regression, the coefficient on the underlying innovation declines from -0.47 to -0.21 percentage points, while the implied-price-impact coefficient remains economically large at -0.64 percentage points. Adding predictable household excess demand leaves this coefficient essentially unchanged at -0.66 percentage points. Implied price impact therefore contains return information beyond both the magnitude of the newly arriving household demand and the inherited household-demand state.

The combination of a positive contemporaneous coefficient and a negative next-quarter coefficient is consistent with a temporary valuation effect. New household demand is associated with an immediate price increase, but the increase is larger when the institutions holding the stock have limited capacity to accommodate the shock. Part of that initial increase reverses in the following quarter. These estimates remain reduced-form associations rather than causal estimates of market depth. Nevertheless, the use of predetermined ownership and elasticity, the inclusion of the underlying innovation and multiplier main effect, and the stability of the implied-price-impact coefficient support the proposed absorption-capacity mechanism.

4.6 Term Structure of Household-Demand Valuation Wedges

The preceding analysis isolates newly arriving household demand and shows that its immediate pricing consequences depend on the Price Impact Multiplier. I next examine how the return effects associated with the full household excess-demand state evolve across horizons. This distinction is important: innovations capture the arrival of new household demand,

whereas the level of household excess demand captures the valuation effect that remains embedded in the household-demand state.

I define the household-demand-implied valuation wedge as

$$\text{IVW}_{j,t}^H = \text{PIM}_{j,t-1} HED_{j,t} = \frac{HED_{j,t}}{\tilde{\kappa}_{j,t-1}^I}. \quad (46)$$

Unlike $\text{IPI}_{j,t}^H$, which applies the multiplier to the household innovation, $\text{IVW}_{j,t}^H$ applies it to the entire household excess-demand state. It therefore measures the valuation wedge that can remain after the initial household demand arrival.

Figure 2 plots horizon-specific Fama–MacBeth coefficients for household excess demand and the implied valuation wedge. The dependent variable at horizon h is the return realized exactly h quarters after formation rather than the cumulative return through quarter h . Both explanatory variables are standardized within quarter, making their coefficient magnitudes directly comparable.

Both measures predict lower subsequent returns, but the implied valuation wedge produces substantially stronger predictability. At the one-quarter horizon, a one-standard-deviation increase in household excess demand predicts a 0.42-percentage-point lower return. The corresponding coefficient for the implied valuation wedge is approximately -1.00 percentage point. The difference remains economically large at later horizons, showing that accounting for institutional absorption capacity materially strengthens the identification of demand-induced valuation effects.

The return correction unfolds gradually. The implied valuation wedge predicts significantly lower returns throughout the first three years and remains significant at several later horizons. Its coefficient declines from approximately -1.00 percentage point in the following quarter to -0.68 percentage points after one year and -0.35 percentage points after three years. The gradual decline indicates that the return consequences of household excess demand do not disappear immediately after formation. Instead, the valuation effect remains

in prices and corrects over successive quarters as household excess demand dissipates.

Overall, the term structure supports the joint role of demand persistence and institutional absorption capacity. The household innovation determines the initial demand arrival, the Price Impact Multiplier governs the magnitude of its price effect, and the persistent household excess-demand state determines how long that effect remains embedded in prices. The economically relevant object is therefore household demand relative to the elastic institutional capital available to absorb it.

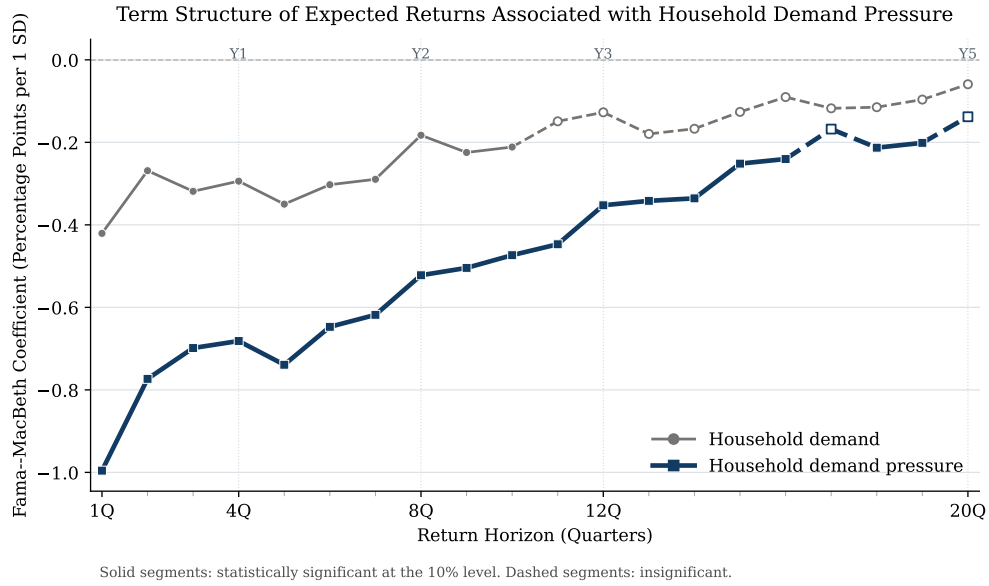


Figure 2: Term Structure of Future Returns Associated with Household Excess Demand and the Implied Valuation Wedge

Notes: This figure plots horizon-specific Fama–MacBeth coefficients from regressions of future quarterly stock returns on household excess demand and the household-demand-implied valuation wedge. The implied valuation wedge is

$$IVW_{j,t}^H = \text{PIM}_{j,t-1} HED_{j,t} = \frac{HED_{j,t}}{\tilde{\kappa}_{j,t-1}^I},$$

where $\text{PIM}_{j,t-1}$ is the Price Impact Multiplier and $\tilde{\kappa}_{j,t-1}^I$ is regularized lagged institutional absorption capacity. The dependent variable at horizon h is the stock return realized exactly h quarters after formation. Both explanatory variables are standardized within quarter, so coefficients represent percentage-point return differences associated with a one-standard-deviation increase in the indicated measure. Solid line segments denote estimates significant at the 5% level, while dashed segments denote insignificant estimates. The implied valuation wedge produces larger and more persistent negative future-return associations than household excess demand alone.

4.7 Portfolio Evidence

I complement the Fama–MacBeth evidence with portfolio tests. Each quarter, stocks are sorted separately into deciles based on the household excess-demand state and household-demand-implied price impact. Decile 1 contains stocks with the highest value of the sorting signal, while Decile 10 contains stocks with the lowest value. The reported portfolio is D1–D10. A negative alpha therefore indicates that stocks with high household excess demand or high implied price impact subsequently underperform stocks with low values of the corresponding signal. Six-factor alphas are estimated relative to the market, SMB, HML, RMW, CMA, and momentum factors.

Table 6 reports the results. The household excess-demand state contains economically important longer-horizon return information. In the full sample, the equal-weighted D1–D10 portfolio earns a next-quarter alpha of -1.76% and a next-year alpha of -4.36% . The next-quarter value-weighted alpha is small and insignificant, but the next-year value-weighted alpha is -3.90% and statistically significant. The annual value-weighted effect is also present among large firms, for which the alpha equals -4.47% . Thus, the longer-horizon return information in the household excess-demand state is not confined to equal-weighted portfolios or microcapitalization stocks.

Household-demand-implied price impact produces its strongest abnormal returns over the following quarter. The full-sample equal-weighted D1–D10 portfolio earns a next-quarter alpha of -2.27% . Among small firms, the value-weighted next-quarter alpha is -1.36% , compared with an insignificant 0.28% among large firms. The concentration among small firms is consistent with, but does not by itself establish, the absorption-capacity mechanism because firm size is only an indirect proxy for the availability of elastic institutional capital. The direct evidence for that mechanism comes from the Price Impact Multiplier tests reported in the preceding subsection.

Taken together, the portfolio results reinforce the horizon-specific evidence from the Fama–MacBeth regressions. Household-demand-implied price impact identifies sharp short-

term reversals following newly arriving household demand, whereas the household excess-demand state contains additional longer-horizon return information. Expressed in the opposite low-minus-high direction, the highlighted portfolios earn six-factor alpha spreads of 1.36% over the subsequent quarter and 3.90% over the subsequent year.

Table 6: Six-Factor Alphas of Household Excess-Demand and Implied-Price-Impact Portfolios

Signal	Universe	Weighting	Next quarter		Next year	
			FF6 alpha	<i>t</i> -statistic	FF6 alpha	<i>t</i> -statistic
Household excess-demand state	All firms	EW	−1.76***	(−4.48)	−4.36**	(−2.58)
		VW	−0.40	(−1.07)	−3.90***	(−3.16)
	Small firms	EW	−4.57***	(−7.11)	−16.42***	(−5.74)
		VW	−1.09*	(−1.90)	−3.60	(−1.50)
	Large firms	EW	0.15	(0.60)	−2.19*	(−1.69)
		VW	0.49	(1.18)	−4.47***	(−3.13)
Household-demand-implied price impact	All firms	EW	−2.27***	(−5.08)	−6.37***	(−3.68)
		VW	−0.13	(−0.27)	1.38	(0.81)
	Small firms	EW	−3.85***	(−6.37)	−10.40***	(−4.55)
		VW	−1.36***	(−3.97)	−2.55**	(−2.59)
	Large firms	EW	0.34	(0.97)	1.09	(0.98)
		VW	0.28	(0.55)	3.50*	(1.95)
Formation quarters			149		149	

Each quarter, firms are sorted into deciles based separately on the household excess-demand state and household-demand-implied price impact within the indicated universe. D1 contains firms with the highest signal, while D10 contains firms with the lowest signal. The reported zero-cost portfolio is D1–D10. A negative alpha therefore indicates that stocks with high household excess demand or high implied price impact subsequently underperform stocks with low values of the corresponding signal.

Household excess demand, $HED_{j,t}$, is the excess-demand measure for the residual non-13F household sector. Household-demand-implied price impact is

$$IPI_{j,t}^H = s_{j,t-1}^H \nu_{j,t}^H \text{PIM}_{j,t-1},$$

where $\nu_{j,t}^H$ is the household excess-demand innovation, $s_{j,t-1}^H$ is lagged household ownership, and $\text{PIM}_{j,t-1}$ is the inverse of regularized lagged institutional absorption capacity.

Small firms have log market equity less than or equal to the within-quarter median, while large firms are above the median. EW and VW denote equal- and value-weighted portfolio returns, respectively. Value weights use market equity measured at portfolio formation.

FF6 alpha is the intercept from a time-series regression on the market, SMB, HML, RMW, CMA, and momentum factors. Next-quarter alpha uses portfolio and factor returns realized during quarter $t + 1$. Next-year alpha is estimated directly using cumulative portfolio returns from $t + 1$ through $t + 4$ and factors compounded over the same horizon; it is not obtained by compounding the next-quarter alpha.

Alphas are reported in percent. Parentheses report Newey–West *t*-statistics with four lags. Significance levels are *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

5 Additional Evidence and Robustness Tests

This section examines alternative interpretations of the main results and provides additional evidence on the underlying mechanism. I first test whether household and institutional excess demand differ in their information about future firm fundamentals and earnings news. I then examine ownership dynamics following household demand innovations and conduct additional robustness tests of the Price Impact Multiplier.

5.1 Fundamental Information Channel

The main results reveal a sharp difference between household and institutional excess demand. Innovations in both sectors are positively associated with contemporaneous returns, but only household excess demand predicts a persistent subsequent reversal. One potential explanation is that the two sectors differ in the fundamental information contained in their demand.

In models of informed trading, demand moves prices because it reveals information about future cash flows. The resulting price adjustment need not subsequently reverse (Kyle, 1985a; Llorente et al., 2002). By contrast, demand arising from sentiment, liquidity needs, or other non-informational motives can temporarily move prices away from fundamental values and generate subsequent return reversals (De Long et al., 1990b; Llorente et al., 2002). I examine this distinction using future accounting outcomes and returns around subsequent earnings announcements.

For each sector $s \in \{H, I\}$, I decompose excess demand as

$$ED_{j,t}^s = \hat{\alpha}_t^s + \widehat{ED}_{j,t}^{s,P} + \nu_{j,t}^s, \quad \widehat{ED}_{j,t}^{s,P} = \hat{\rho}_t^s ED_{j,t-1}^s, \quad (47)$$

where $\widehat{ED}_{j,t}^{s,P}$ is the predictable component and $\nu_{j,t}^s$ is the innovation. The predictable component captures the persistent excess-demand state inherited from the previous quarter, while the innovation captures newly arriving excess demand.

I estimate horizon-specific Fama–MacBeth regressions in which all four household and institutional components enter jointly:

$$Y_{j,t+h} = a_{t,h} + \beta_{t,h}^{H,P} \widehat{ED}_{j,t}^{H,P} + \beta_{t,h}^{H,N} \nu_{j,t}^H + \beta_{t,h}^{I,P} \widehat{ED}_{j,t}^{I,P} + \beta_{t,h}^{I,N} \nu_{j,t}^I + \Gamma'_{t,h} X_{j,t} + u_{j,t+h}. \quad (48)$$

The first set of dependent variables consists of profitability and asset growth measured one and four quarters after formation. The controls include current profitability, asset growth, dividend payout, log market equity, and momentum. These regressions therefore ask whether excess demand predicts future fundamentals conditional on their current values and the remaining firm characteristics.

Future accounting outcomes may nevertheless reflect information that was already publicly available at formation. I therefore complement the fundamental regressions with an earnings-announcement test. For each firm-quarter, I identify the first subsequent I/B/E/S actual-EPS release and construct the two-day market-adjusted return

$$CAR_{j,[0,+1]} = \sum_{d=0}^1 (r_{j,\tau+d} - r_{\tau+d}^M), \quad (49)$$

where τ is the first CRSP trading day on or after the announcement date and r^M is the CRSP value-weighted market return. I then re-estimate equation (48) using the announcement return as the dependent variable.

Table 7 reports the results. Predictable household excess demand forecasts lower future profitability but greater future asset growth. A one-standard-deviation increase in predictable household excess demand is associated with profitability that is 0.18 percentage points lower in the following quarter and 0.85 percentage points lower after one year. At the same time, asset growth is 0.81 percentage points higher in the following quarter and 1.66 percentage points higher after one year. Thus, the persistent household excess-demand state is concentrated in firms that subsequently expand but become less profitable.

Institutional excess demand exhibits a different relation with future profitability. Both predictable institutional excess demand and institutional innovations forecast higher profitability. The coefficient on predictable institutional excess demand increases from 0.23 percentage points in the following quarter to 0.48 percentage points after one year. The corresponding coefficients on institutional innovations are 0.23 and 0.73 percentage points. Institutional excess demand therefore contains information about subsequent operating performance beyond that contained in current fundamentals and the remaining controls.

The earnings-announcement results provide a more direct distinction between the sectors. Predictable institutional excess demand forecasts a 0.06-percentage-point higher abnormal return around the next earnings announcement, whereas predictable household excess demand forecasts a 0.12-percentage-point lower announcement return. The coefficients on the innovation components are smaller and statistically weaker. The information difference is therefore concentrated primarily in the predictable excess-demand states: institutions remain overweight firms that subsequently release favorable earnings news, while persistent household excess demand is followed by unfavorable news.

Taken together, the accounting and announcement evidence helps explain why institutional excess demand does not generate the long-horizon reversal observed for households. Institutional excess demand is positively associated with subsequent profitability and earnings news, making its contemporaneous price effect more likely to reflect information about fundamental value. Household excess demand instead identifies firms that expand but subsequently experience weaker profitability and adverse earnings news. These tests remain associational and do not establish that institutions possess private information or that household investors behave irrationally. They nevertheless show that the sector-level return differences cannot be explained solely by differences in excess-demand persistence.

Table 7: Excess-Demand Components, Future Fundamentals, and Earnings News

Excess-demand component	Future profitability		Future asset growth		Earnings-announcement
	$t + 1$	$t + 4$	$t + 1$	$t + 4$	CAR[0, +1]
Predictable household excess demand	−0.0018*** (−5.24)	−0.0085*** (−7.94)	0.0081*** (16.67)	0.0166*** (13.34)	−0.12*** (−5.46)
Household excess-demand innovation	−0.0024*** (−6.26)	0.0053*** (6.26)	−0.0014*** (−4.38)	0.0124*** (15.18)	−0.02* (−1.73)
Predictable institutional excess demand	0.0023*** (8.01)	0.0048*** (8.18)	0.0020*** (7.12)	0.0030*** (5.06)	0.06** (2.46)
Institutional excess-demand innovation	0.0023*** (6.84)	0.0073*** (11.94)	−0.0006** (−2.60)	0.0042*** (10.81)	0.02 (1.61)
Controls	Yes	Yes	Yes	Yes	Yes
Formation quarters	150	147	150	147	151
Average cross-sectional N	3,675	3,407	3,675	3,407	2,403

This table reports Fama–MacBeth regressions of future fundamentals and subsequent earnings-announcement returns on the predictable and innovation components of household and institutional excess demand. All four components enter each regression jointly and are standardized independently within the formation quarter.

The predictable component is $\hat{\rho}_t^s ED_{j,t-1}^s$ from the quarter-specific projection of current sector excess demand on its consecutive one-quarter lag. The innovation is the corresponding regression residual. Profitability is operating profit divided by book equity, and asset growth is gat . Fundamental regressions include current profitability, asset growth, dividend payout, log market equity, and momentum. Each regression therefore conditions on the current value of its dependent variable.

CAR[0, +1] is the two-day market-adjusted return around the first I/B/E/S actual-EPS release following formation. The event window begins on the first CRSP trading day on or after the I/B/E/S release date. Abnormal returns are daily stock returns minus the CRSP value-weighted market return. CAR coefficients are reported in percentage points; profitability and asset-growth coefficients are reported in their underlying ratio units.

Parentheses report Newey–West t -statistics with four lags. Significance levels are *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

5.2 Ownership Reallocation Following High Household Demand

The return results show that a high household-demand-implied valuation wedge is associated with lower subsequent returns. This interpretation requires the estimated household-demand state to correspond to economically meaningful ownership reallocation rather than merely statistical variation in the demand system. I therefore examine how household and institutional ownership change when a stock first enters a high-implied-valuation-wedge state and how those positions evolve afterward.

The household-demand-implied valuation wedge is

$$IVW_{j,t}^H = PIM_{j,t-1} HED_{j,t} = \frac{HED_{j,t}}{\tilde{\kappa}_{j,t-1}^I}. \quad (50)$$

I define a high-wedge onset as a firm-quarter in which $IVW_{j,t}^H$ enters the top within-quarter quintile after not belonging to that quintile in quarter $t - 1$. This definition identifies the beginning of an unusually high household-demand state rather than repeatedly classifying firms whose valuation wedge remains persistently high.

For each investor group τ , I estimate

$$\Delta Own_{j,t \rightarrow t+h}^\tau = a_{t,h} + \beta_h^\tau HighIVWOnset_{j,t} + \Gamma_h' X_{j,t} + u_{j,t+h}^\tau, \quad (51)$$

where $X_{j,t}$ contains profitability, asset growth, dividend payout, log market equity, and momentum. For the formation-quarter test, the dependent variable is the ownership change from $t - 1$ to t . For the subsequent reallocation tests, it is the cumulative ownership change from t to $t + 4$ or $t + 12$. The coefficient β_h^τ therefore measures the difference in the ownership change of investor group τ between firms entering a high-wedge state and other firms in the same formation quarter.

Table 8 reports the results. During the formation quarter, household ownership increases by 1.01 percentage points more for high-wedge-onset firms than for other firms. Because

household ownership is measured as shares outstanding minus aggregate reported 13F ownership, this increase has an accounting counterpart in lower aggregate institutional ownership. The investor-type estimates identify the composition of the institutional reduction. Mutual funds reduce ownership by 0.45 percentage points, investment advisors by 0.26 percentage points, and banks by 0.19 percentage points. Insurance companies, pension funds, and other institutions make smaller reductions. Thus, the onset of a high household valuation wedge coincides with households accumulating ownership while institutions reduce theirs.

The direction of reallocation reverses after formation. Over the following year, household ownership decreases by 2.08 percentage points more among high-wedge-onset firms, while every major institutional category increases its ownership. Mutual funds and investment advisors exhibit the largest subsequent increases, at 0.84 and 0.70 percentage points, respectively, followed by banks at 0.36 percentage points. The transfer continues over longer horizons. Three years after formation, household ownership has declined by 4.07 percentage points relative to other firms, while mutual funds, investment advisors, and banks have increased ownership by 1.70, 1.28, and 0.65 percentage points, respectively.

These findings provide ownership-based support for the mechanism. High household-demand valuation wedges coincide with household accumulation and institutional reductions during the formation quarter. Households then gradually unwind their positions, while institutions progressively increase ownership. The multi-year rotation is consistent with the gradual return correction documented in the main analysis: household positions do not disappear immediately, and institutional absorption occurs over multiple quarters rather than entirely during the formation period.

The evidence should nevertheless be interpreted as quarterly ownership reallocation rather than direct identification of trade initiation. Holdings are observed only at quarter-end, and household ownership is inferred as the residual of shares outstanding net of reported 13F positions. The analysis cannot recover intraperiod order submission or establish that households causally initiate every transaction. It does show that the household-demand

measure corresponds to economically meaningful ownership changes and that the onset of a high implied valuation wedge is followed by a systematic rotation from residual household ownership toward institutional ownership.

Table 8: Ownership Reallocation Around High Household-Demand Valuation Wedges

Investor group	Formation-quarter change	Subsequent one-year change	Subsequent three-year change
	1.01*** (3.29)	−2.08*** (−12.24)	−4.07*** (−13.13)
Household/outside sector	−0.19*** (−5.19)	0.36*** (7.99)	0.65*** (7.77)
Banks	−0.05*** (−4.16)	0.10*** (5.89)	0.24*** (5.74)
Insurance companies	−0.26*** (−2.66)	0.70*** (9.11)	1.28*** (8.37)
Investment advisors	−0.45*** (−3.03)	0.84*** (7.72)	1.70*** (8.21)
Mutual funds	−0.04*** (−3.62)	0.04*** (2.91)	0.10*** (4.52)
Pension funds	−0.02* (−1.78)	0.04* (1.88)	0.08*** (3.76)
Other institutions			
Controls	Yes	Yes	Yes
Formation quarters	151	151	151
Average cross-sectional N	3,789	3,403	2,864

This table examines ownership reallocation around the onset of a high household-demand-implied valuation wedge:

$$IVW_{j,t}^H = \frac{HED_{j,t}}{\max\{\kappa_{j,t-1}^I, Q_{0.05,t-1}^+(\kappa^I)\}}.$$

The measure is winsorized at the first and 99th percentiles within quarter. A high-wedge onset occurs when a firm enters the top within-quarter quintile at t after not belonging to that quintile at $t - 1$.

Each cell reports $\hat{\beta}_h^\tau$ from equation (51). Formation-quarter change is the change in investor group τ 's ownership share from $t - 1$ to t . Subsequent one- and three-year changes are measured from t to $t + 4$ and from t to $t + 12$, respectively. Coefficients are percentage points of shares outstanding. A positive coefficient indicates that the investor group increases ownership more around high-wedge onsets than for other firms in the same formation quarter.

Controls include profitability, asset growth, dividend payout, log market equity, and momentum measured at formation. Missing ownership for an institutional type is set to zero only when the corresponding firm-quarter is otherwise observed. Household/outside ownership is manager-zero residual ownership and includes non-13F investors and pooled small institutions. It should not be interpreted as a directly observed pure retail portfolio.

Parentheses report Newey–West t -statistics with four lags. Significance levels are *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

5.3 Price Impact Multiplier versus Conventional Limits to Arbitrage

The main results show that institutional absorption capacity amplifies the return consequences of household excess demand. A natural concern is that the Price Impact Multiplier may simply repackage conventional limits-to-arbitrage characteristics. Stocks with a high multiplier may also be smaller, have greater idiosyncratic volatility, or receive less analyst coverage. Each characteristic can independently increase the persistence of a demand-induced valuation effect.

The Price Impact Multiplier nevertheless differs conceptually from these proxies. Firm size, idiosyncratic volatility, and analyst coverage capture trading costs, arbitrage risk, and the information environment. The Price Impact Multiplier instead measures the inverse of the elastic institutional capital available to absorb household demand. The relevant empirical question is therefore whether the multiplier contains information about household- demand return reversals beyond these conventional firm characteristics.

For the robustness tests, I define

$$HighPIM_{j,t} = \mathbb{1} \{PIM_{j,t-1} > \text{Median}_t(PIM_{j,t-1})\}. \quad (52)$$

Because the multiplier is the inverse of institutional absorption capacity, this classification is equivalent to identifying stocks with below-median lagged institutional capacity. The binary indicator is used only to make the interaction coefficients directly comparable with conventional binary limits-to-arbitrage proxies; the main analysis continues to use the continuous Price Impact Multiplier.

I conduct two tests. First, I estimate the High-PIM interaction separately within small and large firms, high- and low-IVOL firms, and low- and high-coverage firms. Within each

subsample, I estimate

$$\begin{aligned}
R_{j,t \rightarrow t+h} = & a_{t,h} + \beta_h HED_{j,t} + \gamma_h HighPIM_{j,t} \\
& + \delta_h (HED_{j,t} \times HighPIM_{j,t}) + \Gamma'_h X_{j,t} + u_{j,t+h}.
\end{aligned} \tag{53}$$

The coefficient δ_h measures whether household excess demand predicts a larger return reversal among High-PIM stocks within a given size, IVOL, or analyst-coverage group.

Second, I estimate a joint horse race on the common sample:

$$\begin{aligned}
R_{j,t \rightarrow t+h} = & a_{t,h} + \beta_h HED_{j,t} + \delta_h (HED_{j,t} \times HighPIM_{j,t}) \\
& + \sum_k \left[\gamma_{k,h} Z_{j,t}^k + \theta_{k,h} (HED_{j,t} \times Z_{j,t}^k) \right] \\
& + \gamma_h^{PIM} HighPIM_{j,t} + \Gamma'_h X_{j,t} + u_{j,t+h},
\end{aligned} \tag{54}$$

where

$$Z_{j,t}^k \in \{Small_{j,t}, HighIVOL_{j,t}, LowCoverage_{j,t}\}.$$

Household excess demand, all indicator main effects, and all four interaction terms enter simultaneously. In this specification, δ_h measures the incremental difference in the household-demand return slope associated with a high Price Impact Multiplier after holding the conventional interactions fixed.

Panel A of Table 9 shows that the High-PIM result is not confined to small firms. Among small firms, the interaction is -2.31 percentage points over the following quarter and -21.91 percentage points over three years. The effect is smaller but remains statistically significant among large firms, at -0.64 percentage points over the following quarter and -9.88 percentage points over three years. The Price Impact Multiplier therefore contains information about household-demand reversals after holding the broad size category fixed.

The IVOL results indicate that the multiplier and conventional arbitrage risk are complementary. Among high-IVOL firms, the High-PIM interaction is large and statistically significant at every horizon, reaching -22.23 percentage points over three years. Among

Table 9: Price Impact Multiplier and Conventional Limits to Arbitrage

Test or interaction	1Q	1Y	3Y	5Y
<i>Panel A: High-PIM interaction within limits-to-arbitrage groups</i>				
	−2.31***	−7.49***	−21.91***	−29.70***
Small firms	(−7.86)	(−5.79)	(−6.61)	(−3.94)
	−0.64***	−3.01***	−9.88***	−11.73***
Large firms	(−2.73)	(−3.81)	(−4.74)	(−3.44)
	−2.29***	−8.15***	−22.23***	−35.73***
High IVOL	(−7.78)	(−6.30)	(−7.19)	(−5.65)
	−0.18	−0.79	−4.47***	−6.54**
Low IVOL	(−0.87)	(−1.30)	(−3.09)	(−2.58)
	−2.05***	−7.17***	−19.29***	−30.05***
Low analyst coverage	(−9.23)	(−8.65)	(−9.08)	(−8.07)
	−0.40	−1.80	−5.43**	−6.01
High analyst coverage	(−1.24)	(−1.36)	(−2.09)	(−1.27)
<i>Panel B: Joint horse race on the common sample</i>				
	−1.20***	−4.34***	−12.47***	−16.46***
Household excess demand × High PIM	(−5.35)	(−5.07)	(−6.04)	(−4.21)
	−0.61***	−2.18**	−6.18***	−14.14***
Household excess demand × Small firm	(−3.02)	(−2.27)	(−3.81)	(−3.75)
	−0.68***	−2.61***	−7.56***	−10.84***
Household excess demand × High IVOL	(−5.34)	(−6.69)	(−6.19)	(−5.05)
	−0.36*	−1.35***	−3.92***	−6.68***
Household excess demand × Low coverage	(−1.90)	(−4.43)	(−3.96)	(−3.90)
All constituent main effects	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Common-sample quarters	145	145	145	145
Common-sample average N	3,783	3,544	2,996	2,569

This table examines whether the Price Impact Multiplier contains information beyond conventional limits-to-arbitrage proxies. The dependent variable is the cumulative raw stock return from quarter $t + 1$ through the indicated horizon. Household excess demand is standardized within quarter.

Panel A restricts the sample to the indicated group and reports the coefficient on household excess demand interacted with High PIM. Each regression also includes household excess demand, the High-PIM indicator, and their interaction. High PIM is defined within quarter before the sample is divided into size, IVOL, or analyst-coverage groups. A negative coefficient indicates that household excess demand predicts a larger return reversal among High-PIM stocks within that group.

Panel B includes household excess demand, High PIM, small size, high IVOL, low analyst coverage, all four interactions with household excess demand, and all constituent main effects simultaneously. Each interaction coefficient therefore measures incremental heterogeneity in the household-demand return relation conditional on the remaining proxies.

High PIM identifies stocks with above-median lagged Price Impact Multipliers, equivalently below-median lagged institutional absorption capacity. Small firms have below-median log market equity. High-IVOL firms have above-median lagged 12-month Fama–French five-factor residual volatility. Low coverage includes zero-coverage firms and covered firms at or below the within-quarter median among firms with positive analyst coverage. All classifications are formed within quarter.

Controls are profitability, asset growth, dividend payout, log market equity, and momentum measured at formation quarter t . Entries are percentage-point Fama–MacBeth coefficients. Parentheses report Newey–West t -statistics with four lags. Quarterly cross sections in which the full joint design matrix is rank deficient are excluded from Panel B. Significance levels are *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

low-IVOL firms, the interaction is insignificant over the following quarter and year but becomes significantly negative at the three- and five-year horizons. Limited institutional absorption capacity therefore has its strongest short-run consequences when arbitrageurs also face substantial idiosyncratic risk, although a longer-run relation remains among low-IVOL stocks.

A similar pattern appears for analyst coverage. High PIM strongly amplifies the return reversal among low-coverage firms at every horizon. Among high-coverage firms, the interaction is smaller and generally insignificant, except at the three-year horizon. Information production may therefore facilitate the identification and correction of household-demand-induced valuation effects. More importantly, the High-PIM result within the low-coverage group shows that analyst coverage does not subsume the information contained in institutional absorption capacity.

Panel B provides the more demanding test. When all four interactions enter simultaneously, the High-PIM interaction remains negative and statistically significant at every horizon. Its coefficient is -1.20 percentage points over the following quarter, -4.34 percentage points over one year, -12.47 percentage points over three years, and -16.46 percentage points over five years. It is also the largest interaction in absolute magnitude at each horizon.

The conventional proxies remain significant, indicating that they capture additional frictions affecting the correction of household-demand-induced valuation effects. The evidence therefore does not imply that the Price Impact Multiplier subsumes conventional limits to arbitrage. Rather, the multiplier provides incremental information about the elastic institutional capital available to accommodate household excess demand, while size, IVOL, and analyst coverage capture complementary trading, risk, and information frictions. These reduced-form results support the Price Impact Multiplier as a distinct empirical dimension of the proposed market-clearing mechanism, but they should not be interpreted as establishing an independent causal effect.

6 Conclusion

This paper studies how household demand affects stock prices across horizons and why household shocks of similar magnitude can generate different pricing outcomes. Using investor-level demand systems, I estimate household excess demand as the component of household portfolio demand not explained by instrumented prices or observed firm characteristics. I then separate newly arriving household demand from the predictable excess-demand state inherited from previous quarters.

To quantify the role of institutional absorption capacity, I construct a Price Impact Multiplier as the inverse of the elasticity-weighted institutional capital holding each stock. Multiplying a household demand innovation by this multiplier produces its household-demand-implied price impact. Conditional on the underlying innovation, a one-standard-deviation increase in implied price impact is associated with a 6.44-percentage-point higher contemporaneous return and a 0.64-percentage-point lower next-quarter return. A value-weighted high-minus-low portfolio formed among small firms on implied price impact earns a six-factor alpha of -1.36% over the following quarter.

Household demand also has persistent return consequences. Household excess demand has a quarterly persistence coefficient of approximately 0.88, and stocks with high household excess demand earn lower returns for nearly three years. Ownership dynamics show that households accumulate ownership when a high household-demand valuation wedge emerges and subsequently unwind those positions as institutions gradually increase their ownership. Consistent with this slow adjustment, a full-sample value-weighted high-minus-low portfolio sorted on the household excess-demand state earns a six-factor alpha of -3.90% over the following year.

The Price Impact Multiplier continues to identify stronger household-demand reversals after controlling for firm size, idiosyncratic volatility, and analyst coverage. Additional evidence indicates that institutional excess demand is more positively associated with future profitability and earnings news, helping explain why institutional demand does not generate

the same persistent reversal. Overall, newly arriving household demand produces short-term valuation effects whose magnitude depends on institutional absorption capacity, while the gradual dissipation of the household excess-demand state extends those effects across longer horizons.

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